Asymmetric Price Competition on Hybrid Platforms: Theory and Evidence from Amazon*

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Abstract

We study how hybrid platform operation, in which a platform competes with third-party sellers while charging a commission on their sales, shapes pricing incentives. Our model identifies two channels induced by the ad valorem commission – a revenue-sharing and a double-markup effect – that generate asymmetric pricing behavior between the platform and third-party sellers. Using data from Amazon's U.S. marketplace, we find patterns consistent with the model's predictions: third-party prices respond modestly more strongly to third-party entry and exit than to Amazon entry and exit, and significantly more strongly than Amazon prices do to common commodity-cost shocks.

Keywords: E-Commerce, Hybrid Platform, Online Pricing, Pass-Through

JEL Classification: D4, L1, E3

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1 Introduction

A growing number of e-commerce platforms – from Amazon and Walmart to Zalando and JD.com – operate in *hybrid* mode: they run marketplaces that connect third-party (3P) sellers with consumers, while also acting as retailers and selling products directly to consumers. Similar arrangements arise in other digital markets, including mobile app stores (e.g., Apple's App Store and Google Play) and major gaming-console storefronts (e.g., the PlayStation Store, Nintendo eShop, and Xbox Store). Another key feature of these platforms is that, on the marketplace side, they typically charge business users an ad valorem commission on sales. Commission rates can be substantial, reaching as high as 30% in some digital marketplaces.

Hybrid operation therefore creates an unusual competitive relationship: the platform competes directly with 3P sellers while simultaneously earning a share of their revenues.[1] Such an arrangement has raised antitrust concerns about an uneven playing field on the marketplace (EC Vertical Guidelines, 2022). In its 2021 Statement of Objections against Apple, the European Commission expressed the preliminary view that Apple's rules distorted competition by raising the costs of competing music-streaming providers, thereby leading to higher prices for consumers (EC, 2021). In addition, competition authorities in Europe and the United States have scrutinized or challenged various aspects of Amazon's conduct in competing with 3P sellers, including self-preferencing, the use of non-public seller data, and pricing practices that may constrain or deter price competition by marketplace sellers (EC, 2022; CMA, 2023; FTC, 2024; Bundeskartellamt, 2026).

In this paper, we seek to understand how the distinctive institutional structure of hybrid platforms shapes price competition in downstream product markets, both theoretically and empirically. Rather than focusing on the exclusionary or discriminatory conduct highlighted in current antitrust debates, we ask a more fundamental but less studied question: how does hybrid operation, combined with revenue sharing, influence downstream pricing incentives and competitive outcomes in its own right?

To address this question, we first develop a model of price competition on a hybrid platform that charges an ad valorem commission to 3P sellers, allowing us to systematically investigate the

[1] The setting differs from familiar retail arrangements: unlike traditional retailers, the platform neither takes ownership of 3P sellers' products nor controls their retail prices. This distinguishes the hybrid mode from the case of a retailer selling private-label products alongside branded goods. Moreover, although an ad valorem commission gives the platform a direct stake in 3P sales, the platform does not internalize 3P sellers' costs, distinguishing this arrangement from partial ownership of a 3P seller.

interplay among the pricing forces induced by the commission. On the one hand, the commission inflates 3P sellers' perceived marginal costs, subjecting them to a double-markup distortion that the vertically integrated platform avoids. On the other hand, because the platform earns commission revenue from 3P sales, it internalizes demand diversion from its own product toward 3P products when setting its retail price. Together, these two channels result in asymmetric price competition between 3P sellers and the platform.

This asymmetry gives rise to two key theoretical findings. First, although platform entry intensifies competition relative to no entry, it may exert weaker competitive pressure than entry by an otherwise identical 3P seller. Intuitively, this is because the revenue-sharing effect can offset the downward pricing force associated with the platform's avoidance of the double-markup distortion when the commission rate is relatively low. Second, the platform exhibits lower pass-through of industry-wide cost shocks than 3P sellers. This difference is driven primarily by the double-markup channel, which amplifies the effective cost shock faced by 3P sellers.

Building on these theoretical insights, our second contribution is to provide empirical evidence from the Amazon marketplace on the pricing asymmetries highlighted by the model. We collect a granular panel of prices for tens of thousands of products in narrowly defined product categories sold on `amazon.com`, the U.S.-facing storefront of Amazon. Our data show that Amazon participation is selective and concentrated in high-turnover products, and that Amazon and 3P sellers repeatedly become available and unavailable for a given ASIN in ways that are not readily predictable from observables, and possibly due to stockouts, as in Chen and Tsai (2024). To take our first prediction to the data, we use hundreds of repeated within-product episodes in which sellers temporarily enter or exit as a retailer, and conduct short-run event studies of 3P price responses to entry and exit by Amazon and by other 3P sellers. Under the assumption that the timing of these entry and exit episodes is conditionally exogenous – an assumption for which we provide supporting pre-trend and hazard diagnostics – Amazon entry is followed by a 6% decrease in the prices of 3P sellers selling the same product, while Amazon exit is followed by a 3% increase. Comparing the pricing response of 3P sellers upon 3P entry (exit) to Amazon entry (exit), we find that 3P prices tend to move modestly more strongly following 3P entry or exit than following Amazon entry or exit. These patterns therefore provide suggestive empirical support for the possibility highlighted by our model that platform entry may exert less downward pressure on 3P prices than entry by a 3P seller.

Our second main empirical finding is a seller-type asymmetry in cost pass-through: Amazon's posted prices respond less than 3P prices to common upstream commodity-price shocks. Using distributed-lag regressions to understand how costs pass through to prices over different time horizons, we estimate how changes in the commodity prices of coffee and cocoa beans affect the prices of ground coffee and baking cocoa sold by 3P sellers and Amazon, respectively. These commodity price changes originate in supply-side shocks, particularly adverse weather and crop disease in coffee- and cocoa-producing regions. Among product-weeks with both seller types active, the estimated commodity-cost pass-through is higher for 3P sellers at every horizon in coffee, with a statistically significant gap at 12, 24, and 48 weeks. The smaller cocoa sample yields the same ordering at every horizon, and a statistically significant gap at 48 weeks in our baseline regression. These patterns persist when we either extend the sample to product-weeks in which both seller types are not necessarily jointly active, or restrict it to narrower subsamples. In particular, the pattern is robust to excluding particularly high-priced products, or to focusing on higher-turnover products. Overall, our evidence is consistent with the asymmetric pass-through of common cost shocks predicted by our model.

At a broader level, our findings speak to the debate over whether hybrid operation should be viewed as inherently anticompetitive. Our results suggest a more nuanced view. Platform entry does increase competitive pressure on incumbent sellers, but this pressure may be weaker than that exerted by an otherwise comparable 3P entrant. At the same time, the platform's lower pass-through of common cost shocks makes its retail price less sensitive to industry-wide cost increases. While this can benefit consumers in the short run, it also means that positive cost shocks place greater upward pressure on 3P prices, potentially shifting demand toward the platform. If such shocks are persistent, or if temporary shifts in market share have lasting effects, this asymmetric exposure could strengthen the platform's competitive position over time.

Related Literature

Our theoretical framework builds on two insights from the literature on agency models (Johnson, 2017; Foros, Kind and Shaffer, 2017; Hu, Zheng and Pan, 2022). First, relative to traditional wholesale models, agency selling leaves downstream pricing and other key retail decisions in the hands of suppliers (Hagiu and Wright, 2015), so that suppliers compete directly in retail prices. Second, under an agency model, the intermediary (e.g., a marketplace) charges suppliers an ad valorem commission, generating a nonstandard double-markup distortion. Existing work, however, does not consider the hybrid setting in which the intermediary also competes downstream with the sellers from whose sales it earns commission revenue. Our model introduces this additional strategic interaction and studies how it reshapes price competition.

A growing literature studies hybrid platforms. The theoretical literature can be organized around three broad questions: the endogenous emergence of the hybrid mode (Etro, 2021; Gautier, Hu and Watanabe, 2023; Hervas-Drane and Shelegia, 2025), its competitive effects (Etro, 2021; Hagiu, Teh and Wright, 2022; Etro, 2023; Anderson and Bedre Defolie, 2024; Hervas-Drane and Shelegia, 2022), and the antitrust implications of this business model, including self-preferencing and imitation (e.g., Hagiu et al., 2022; Choe, Dubus, Matsushima and Shekhar, 2025; Madsen and Vellodi, 2025). Our focus is specifically on how the hybrid mode shapes price competition. Several papers are particularly relevant. Hagiu et al. (2022) show that the platform's own product "price-squeezes" a superior 3P seller, intensifying competition and thus increasing total transactions on the marketplace. In contrast, Anderson and Bedre Defolie (2024) find adverse competitive effects of hybrid operation: the platform optimally raises its commission fee to steer demand toward its own offerings, reducing 3P participation and product variety and ultimately raising final prices. Our paper differs from those in two key aspects. First, these papers endogenize both the platform's choice of business model and its commission fee. In contrast, we view the commission fee as a longer-term choice and therefore treat it as fixed, while allowing the platform to flexibly enter or exit competition with 3P sellers at the product level. This is consistent with the empirical fact in our data that Amazon frequently begins and ceases selling individual products without any corresponding change in the category-level referral fee. Second, by adopting a general oligopoly model of price competition, we can systematically investigate how the interplay between the revenue-sharing channel and the double-markup channel shapes price competition. The homogeneous-preferences and fixed-fee framework of Hagiu et al. (2022) and the monopolistic-competition setting of Anderson and Bedre Defolie (2024) do not generate such strategic pricing interactions. We discuss these differences in detail in Section 3.3.1. Finally, Hervas-Drane and Shelegia (2022) is also closely related to our paper in terms of research question. Like us, the authors study how a platform's stake in rival sellers affects competitive interactions in the marketplace. However, as the authors build on Varian (1980)'s model of sales, their setting and mechanism differ substantially from ours.

Existing empirical literature on hybrid platforms has investigated self-preferencing and steering on hybrid platforms (Lee and Musolff, 2026; Chen and Tsai, 2024; Dendorfer and Seibel, 2026;

Farronato, Fradkin and MacKay, 2023, 2025), Amazon's entry strategy (Zhu and Liu, 2018), and the effects of restricting Amazon's retail activity (Gutiérrez, 2022; Bennati, 2026; Bedre Defolie and Sokullu, 2026). Evidence on how hybrid platform operation shapes within-platform price competition remains limited. Crawford, Courthoud, Seibel and Zuzek (2022) and Korganbekova and Korganbekov (2026) may be most closely related to our paper, as they also study the effects of Amazon entry (as a reseller or under a private-label brand) on 3P merchants. However, they focus primarily on whether Amazon selectively enters successful products pioneered by 3P sellers and study the effects of Amazon entry relative to no entry, whereas we ask whether Amazon's dual role makes the competitive effect of its entry differ from that of an otherwise identical 3P seller. We further study the implications of hybrid operation for asymmetric cost pass-through, which has not been examined in this literature.

Our paper therefore also uncovers a novel implication of hybrid platforms for cost shock transmission. This connects us to the literature on pass-through and incidence under imperfect competition (e.g., Weyl and Fabinger, 2013; Miklos-Thal and Shaffer, 2021). Studying the pass-through of coffee-bean prices, Nakamura and Zerom (2010), Bettendorf and Verboven (2000) and Sangani (2026) are most closely related. Rather than estimating average pass-through, we are, to the best of our knowledge, the first to identify asymmetric cost pass-through as a consequence of hybrid-platform organization and document empirical patterns that are in line with our theory.

Outline. This paper is structured as follows. Section 2 describes the main institutional features of e-commerce platforms like Amazon, introduces the data we have collected, and provides stylized facts on seller participation and prices. Section 3 builds an oligopoly model of price competition on a hybrid platform, and proposes empirical implications. Sections 4 and 5 present empirical results on short-term entry and exit responses and pass-through on the Amazon marketplace, respectively. Section 6 concludes.

2 Institutional Setting and Stylized Facts

2.1 Hybrid Selling on Amazon Marketplace

We focus on the empirical setting of amazon.com, the U.S.-facing storefront of Amazon.[2] Amazon is the world's largest e-commerce platform and has operated in hybrid mode since 2000. 3P sellers account for over 60% of unit sales on Amazon, and 560,000 sellers – predominantly small and medium-sized firms – were active on the U.S. marketplace in 2021 (Federal Trade Commission, 2024).[3] Amazon levies an ad valorem commission (called "referral fee") on each mediated 3P transaction. The referral-fee schedule is organized primarily by product category and revised infrequently;[4] rates are typically between 5 and 20% of the final price. Amazon's first-party retail activity includes private-label offerings (e.g., Amazon Basics), but a major fraction of Amazon's sales in fact concerns non-private-label products that Amazon resells.

Amazon identifies a product by an Amazon Standard Identification Number (ASIN), and aggregates all offers for a given product onto a single detail page. Sellers of the same ASIN compete primarily for the "Buy Box" – the prominent section on the top-right of the page – which captures the vast majority of sales. Its allocation is governed by a proprietary algorithm (Chen, Mislove and Wilson, 2016; Lee and Musolff, 2026; Breslin, 2025). Offers from non-winning sellers are relegated to a less visible list ("Other Sellers on Amazon") below the Buy Box. Sellers can choose to outsource storage and shipping to Amazon ("Fulfilled by Amazon"), with a per-unit fulfillment fee in addition to the ad valorem referral fee, or avoid FBA fees by managing the logistics themselves ("Fulfilled by Merchant").[5] Prior research suggests that seller identity, delivery mode, and Buy Box status make same-ASIN offers imperfect substitutes (Lee and Musolff, 2026), which motivates the use of an oligopoly model to capture price competition.

[2] In the U.S., Amazon likely has a loyal customer base: an estimated 190 million customers are paid subscribers and receive membership benefits; see https://cirpamazon.substack.com/p/updated-us-amazon-prime-member-estimate (accessed 11/02/2026).

[3] Amazon sellers made an average yearly revenue of $250,000 in 2023; see https://sell.amazon.com/blog/amazon-stats (accessed 27/09/2024).

[4] See https://sellercentral.amazon.com/help/hub/reference/external/GTG4BAWSY39Z98Z3 (accessed 27/11/2025); the most recent referral fee change on amazon.com we are aware of occurred in January 2024.

[5] FBA fees depend on product size and weight of the product, not on its price, and hence does not generate the effects that ad valorem fees generate. See https://sellercentral.amazon.com/revcalpublic?ref_=sdus_pri_xscus_rc2&initialSessionID=135-2662106-4270236&ld=NSGoogle_SD_PRI_RC2&pageName=US%3ASD%3AFBA-main&ldStackingCodes=NSGoogle (accessed 19/06/2026).

2.2 Data and Measurement

Data source. We collect data on prices of products sold on Amazon from Keepa (`keepa.com`), a company that provides real-time and historical data on prices, reviews, sales rank, and other time-varying and cross-sectional attributes of products sold on Amazon. Keepa updates its information multiple times per day, and its data have been used in previous academic literature (e.g., He, Reimers and Shiller, 2022; Gutiérrez, 2022; Cabral and Xu, 2021; He, Hollenbeck and Proserpio, 2022; Bennati, 2026)). Our sample period runs from January 2021 to March 2025, and thereby deliberately excludes later periods in which U.S. tariff changes and geopolitical turmoil may have influenced prices and supply chains.

Sample. We focus on subcategories on Amazon that experienced sizable, well-identified upstream cost shocks in recent years and that were not affected by entry of platforms like *Shein* or *Temu*, in particular the ground coffee and cocoa categories (see Section 5).[6] We also collect data for other food (wheat pasta) and non-food subcategories (TVs, washers, T-shirts, USB hubs, and SD cards). Within each product category, at each scrape, we select the 10,000 ASINs with the lowest current "Best Sellers Rank" (hereafter "sales rank") 180-day average sales rank. Because we repeat this procedure over time and retain all selected ASINs, the final sample for a category may contain more than 10,000 ASINs.

Whenever a product is offered on Amazon, we observe the lowest price among 3P offers under each of the delivery modes (FBA and FBM), the lowest price across all offers (including Amazon retail), Amazon's own price, the total number of active sellers, and the seller ID and price of the offer displayed in the Buy Box. Additionally, we observe each product's sales rank. We use the latter to construct variables that identify relatively high-turnover products. For food categories, we recover package sizes (weight in grams) from product titles using the OpenAI API.[7] We then construct a product-day panel using prices recorded at midnight, and aggregate these data to the product-week level by averaging all time-varying variables.

Table 1 reports summary statistics on sample coverage and seller participation for each product

[6] Grocery products now account for a substantial share of Amazon's revenue, suggesting that our categories capture mainstream products. In 2024, Amazon CFO Brian Olsavsky noted "strong growth in items like everyday essentials. This includes items like health, beauty, and personal care, as well as nonperishable grocery," and stated that "the strength in everyday essential's revenue is a positive indicator that customers are turning to us for more of their daily needs." See `https://www.fool.com/earnings/call-transcripts/2024/10/31/amazoncom-amzn-q3-2024-earnings-call-transcript/` (last accessed 01/02/2025).

[7] Manual checks confirm a very high accuracy. In the Appendix, we provide details on how the data were built and also show that food products are sold predominantly in standard household sizes rather than bulk formats.

category. Product variety is substantial in every category: for example, the coffee category alone contains almost 40,000 distinct ASINs, 14,532 of which are in our dataset. Competitive conditions also vary meaningfully across categories. The average number of sellers per ASIN-week ranges from 1.3 for SD cards to 2.5 for coffee and pasta, and the share of ASIN-weeks with at least two observed sellers ranges from 14.3% to 42.0%. Review prevalence also differs markedly across categories. By contrast, products identified as Amazon private-label account for only a very small share of products in every category, never exceeding 0.4% of the products in our sample.

Table 1: Summary of sample by category (ASIN-week panel; 222 weeks)

Category	Products (ASINs)	Avg. seller count (per ASIN-week)	ASIN-weeks with ≥ 2 sellers	With reviews (across ASINs)	Amazon private label (across ASINs)
Cocoa	1,432	2.3	32.4%	67.0%	0.2%
Coffee	14,532	2.5	35.8%	90.6%	0.3%
Pasta	7,709	2.5	42.0%	72.9%	0.4%
Washers	2,167	1.8	21.7%	22.8%	0.0%
TV	6,875	2.3	32.5%	64.8%	0.0%
T-Shirts	22,825	1.5	16.0%	92.4%	0.0%
USB hubs	7,269	2.2	20.0%	43.3%	0.4%
SD cards	12,851	1.3	14.3%	11.4%	0.0%

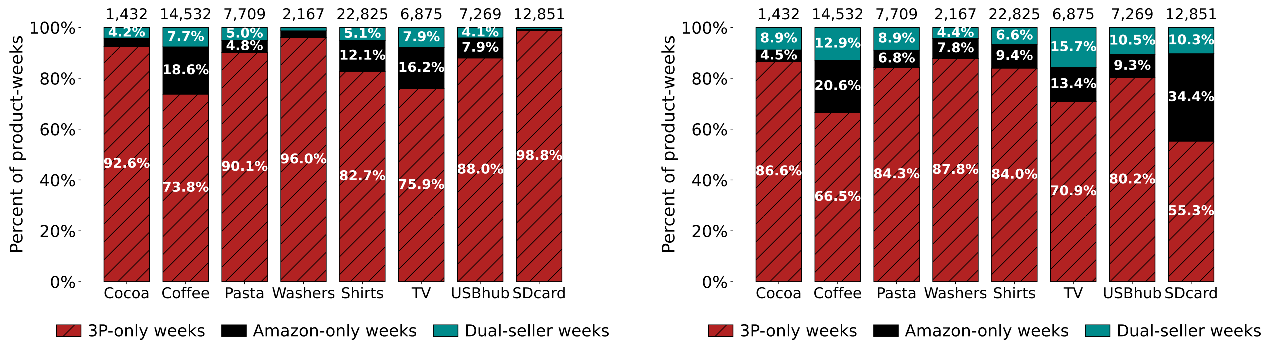
Notes: We track ASINs daily from January 1, 2021 to March 31, 2025 and collapse the data to the ASIN-week level. The average seller count corresponds to the average number of sellers listing an ASIN in a week, computed over ASIN-weeks with at least one observed seller, and includes Amazon retail when present. "With reviews" and "Amazon private label" are product-level shares. The review indicator is measured from the latest observed product page and is therefore time-invariant within the sample.

2.3 Stylized Facts

Fact 1: Amazon retailing is selective and concentrated in high-demand ASINs. Figure 1 displays, for each category, the share of ASIN-week observations that are supplied exclusively by 3P sellers (3P-only), exclusively by Amazon (Amazon-only), or by both (dual-seller). Panel (a) weights each ASIN-week equally and shows that dual-seller states account for between about 0.8% and 7.9% of product-weeks across categories. 3P sellers are the exclusive suppliers in at least 73.8% of product-weeks in every category, while Amazon is the exclusive seller in between 0.4% and 18.6% of product-weeks. Overall, Amazon's retail participation varies markedly across product categories: Amazon is active (either alone or alongside 3P sellers) in roughly 26% of ASIN-weeks in *Coffee* and about 24% in *TVs*, but in less than 2% of ASIN-weeks in *SD Cards*.

Panel (b) of Figure 1 weights ASIN-weeks by their within-category sales-rank percentile, giving more weight to higher-turnover products. Comparing the equal-weighted shares in panel (a) and

Figure 1: Extent of hybrid selling across categories



(a) Equal-weighted product-weeks.

(b) Weighted by inverse of sales-rank percentile.

Notes: Each bar shows the share of ASIN-week observations that are 3P-only, Amazon-only, or dual-seller for a given category. Panel (a) weights each ASIN-week equally. Panel (b) weights ASIN-weeks by the inverse of their within-category sales-rank percentile (1−percentile), giving more weight to higher-turnover products. The sample covers ASINs observed weekly from January 2021 through March 2025. A product-week is classified as 3P-only (Amazon-only) if at least one 3P (Amazon) offer is recorded and the other side is absent; it is classified as dual-seller if both offer the product in the same week. The number of ASINs for each category is displayed above each bar.

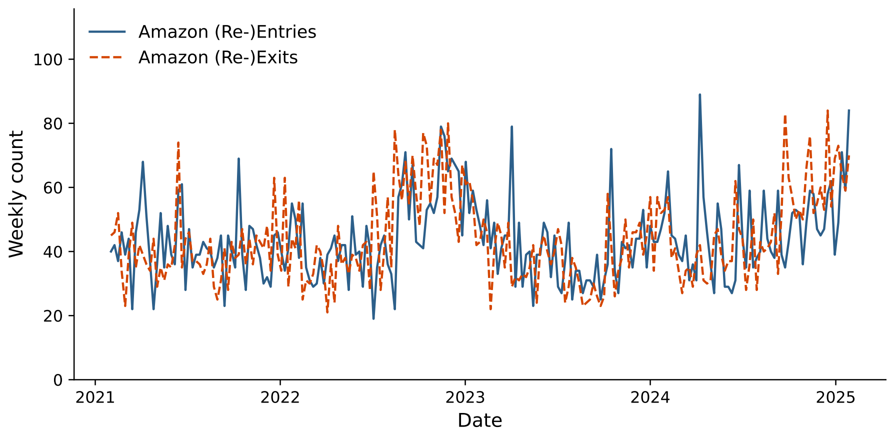
the sales-rank-weighted shares in panel (b) shows that the prevalence of Amazon-only and dual-seller ASIN-weeks increases for all categories except for *Shirts*. For instance, in both *TVs* and *Coffee*, the share of ASIN-weeks in which Amazon is active (Amazon-only or dual-seller) rises from about 24-26% under equal weighting to around 30-34% under sales-rank percentile weighting. For *SD Cards*, Amazon-only and dual-seller states account for around 34% and 10% of sales-rank-weighted ASIN-weeks, respectively. Both shares are thus markedly larger than before. Taken together, these patterns indicate that Amazon's direct retail presence is relatively more common among higher-turnover ASINs, while 3P sellers appear to account for most of the low-turnover "long tail" within categories.

Fact 2: Amazon's retail availability is intermittent and recurrent within ASINs. A central feature of Amazon's own retail activity is that its availability changes repeatedly within the same ASIN. Among ASINs for which Amazon is observed as a retailer before the start of our sample, Amazon frequently moves between periods of presence and absence[8]. Figure 2 plots the weekly numbers of Amazon (re-)entries and (re-)exits for ASINs in the coffee category. These entry and exit events occur throughout the sample period and are not concentrated on a small number of calendar dates. Among ASINs that switch Amazon's presence status, the median ASIN has three Amazon-

[8]We distinguish these recurrent ASIN-level transitions from a small number of coordinated brand-level assortment changes, in which Amazon begins or ceases to offer several ASINs from the same brand at approximately the same time; see Figure C.1 in the Appendix for an example. These episodes are more consistent with deliberate assortment decisions and conceptually distinct from the recurrent within-ASIN variation emphasized here.

present spells and four Amazon-absent spells. Across spells, the median durations are three days when Amazon is present and nine days when it is absent (see Appendix C.2). Similar patterns arise in the other categories. The events therefore mark transitions between recurring spells of Amazon's presence and absence as a reseller, as opposed to permanent product-market entry or exit. While consistent with stockouts or replenishment delays, as discussed by Chen and Tsai (2024), our data do not identify the operational cause of each transition.

Figure 2: Recurring Amazon entries and exits over time



Notes: The figure shows the weekly number of re-entries and re-exits by Amazon over time for the Coffee category. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (which applies to one brand in the case of coffee), as those spikes would reflect systematic brand-level assortment changes rather than product-level entry or exit decisions (see Figure C.1). We only count a re-entry (re-exit) as such if it is followed by a spell of presence (absence) of 5 days or more (the patterns are similar when using other cutoffs).

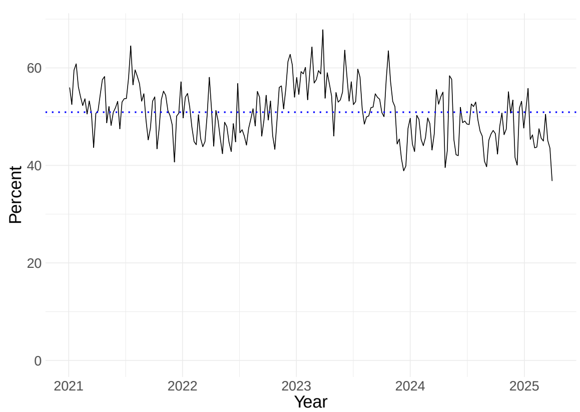
Fact 3: Within ASIN-weeks with both seller types active, Amazon's posted price is often at or below the lowest 3P offer. To avoid conflating observed price differences with product assortment, we restrict attention to ASIN-weeks in which Amazon and at least one 3P seller jointly offer the same product. Figure 3 shows that Amazon's posted price is at or below the lowest observed 3P offer in roughly one-half to two-thirds of these observations across the displayed categories. Fixed-effects comparisons with the lowest overall 3P and lowest FBA offers, reported in Tables C.6–C.9, yield the same pattern, with the clearest negative differences in coffee.[9]

Taken together, these facts show that Amazon's retail arm is a selective competitor: it is more prevalent on high-turnover ASINs, frequently alternates between presence and absence within the same ASIN, and, conditional on direct overlap, Amazon's prices are at or below comparable 3P

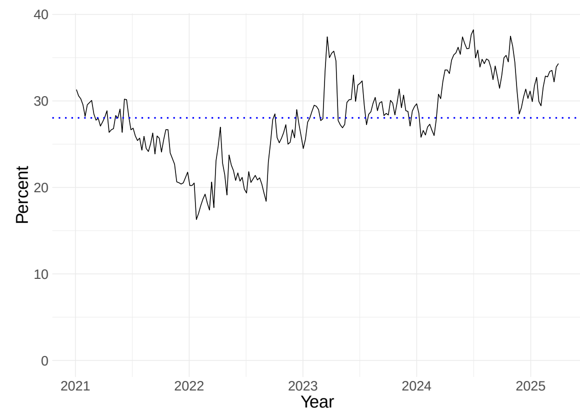
[9] The corresponding estimates relative to FBM offers are negative throughout, but this comparison is less direct because FBM prices include shipping costs.

offers. However, similar price levels need not imply similar pricing incentives. In the next section, we develop a theoretical framework characterizing the pricing incentives of Amazon and 3P sellers.

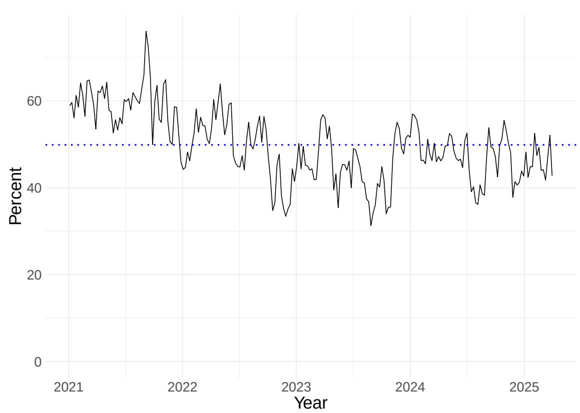
Figure 3: Weekly share of jointly offered ASINs for which Amazon's price exceeds the lowest 3P price



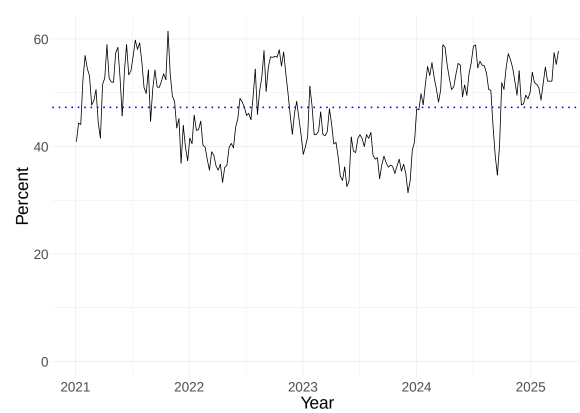
(a) Cocoa.



(b) Coffee.



(c) SD Cards.



(d) USB Hubs.

Notes: Plots use product-week observations in which both a 3P price and Amazon's price are available (i.e., both seller types are active). The blue dotted line indicates the median.

3 Theory

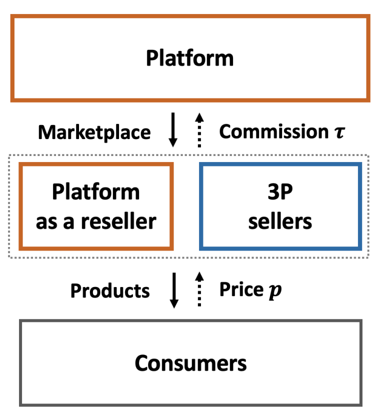
In this section, we develop a model of price competition on a hybrid platform that charges an ad valorem commission to 3P sellers. The platform's dual role as both retailer and intermediary sets it apart from 3P sellers. In this setting, the ad valorem commission shapes price competition through two distinct channels. We then examine how the interplay between these two channels generates asymmetric pricing behavior between the platform and 3P sellers.[10]

3.1 General Framework

Suppose that a monopoly e-commerce platform P operates a marketplace in a given product market. There are n retailers, each selling a differentiated product on the marketplace. The demand faced by retailer $i \in \{1, \ldots, n\}$, denoted $D_i(\mathbf{p}) = D_i(p_i, \mathbf{p}_{-i}) = D_i(p_1, \ldots, p_n)$, depends on its own price p_i and the prices of its rivals $\mathbf{p}_{-i}$. We assume a symmetric demand system. Demand functions are continuously differentiable in prices, with $\partial D_i / \partial p_i < 0$ and $\partial D_i / \partial p_j > 0$ for all $j \neq i$, so that products are substitutes. The marginal cost of retailer $i \in \{1, \ldots, n\}$ is c_i. The platform charges a commission fee $\tau \in (0, 1)$ on the revenue generated by each 3P seller on the platform. Consistent with the institutional setting described in Section 2, in which referral-fee schedules are revised infrequently while Amazon can repeatedly start and stop retailing a given product, we take the commission fee as given and assume that it does not vary with the platform's mode of operation.

We consider two possible modes of platform operation in this product market: the pure marketplace mode and the hybrid mode. In the pure marketplace mode, the platform hosts only 3P products, so all n competing retailers are 3P sellers. In the hybrid mode, the platform not only runs the marketplace but also sells its own product and competes directly with 3P sellers. Figure 4 illustrates the hybrid structure. With n competing retailers, we let the platform be retailer 1 and denote the price and marginal cost of its product by p_P and c_P, respectively, where $p_P = p_1$ and $c_P = c_1$. The remaining retailers $i = 2, \ldots, n$ operate as 3P sellers.

Figure 4: Structure under the hybrid mode



Timing: Under each mode, all retailers simultaneously choose their prices p_i, $i \in \{1, 2, \ldots, n\}$.

[10]All omitted proofs are collected in Appendix A.1.

3.2 Commission-Induced Pricing Forces

We start by characterizing two distinctive channels through which the ad valorem commission affects price competition: the *double-markup channel* in 3P sellers' pricing and the *revenue-sharing channel* in the platform's pricing of its own product. These two channels generate the key pricing forces that shape both equilibrium prices and cost pass-through.

In the pure marketplace mode, each retailer $i \in \{1, 2, \ldots, n\}$ chooses its price p_i to solve

$$\begin{aligned} &\max_{p_i} \left[(1-\tau)p_i - c_i\right] D_i(\mathbf{p}) \\ \Leftrightarrow \quad &\max_{p_i} (1-\tau)\left(p_i - \frac{c_i}{1-\tau}\right) D_i(\mathbf{p}), \end{aligned}$$

where each retailer retains only a fraction $1-\tau$ of its sales revenue while bearing the full marginal cost of the product. Rearranging the profit function, the commission is equivalent to scaling up retailer i's marginal cost by a factor of $1/(1-\tau)$. Following Johnson (2017), we call $\frac{c_i}{1-\tau}$ the retailer's "perceived marginal cost". We use the term *double-markup channel* to refer to the pricing effects generated by τ through inflating 3P sellers' perceived marginal costs. As the first-order condition implies,

$$p_i - \frac{c_i}{1-\tau} = -\frac{D_i(\mathbf{p})}{\partial D_i(\mathbf{p})/\partial p_i}, \qquad i \in \{1, \ldots, n\}, \tag{1}$$

a 3P seller with market power adds another layer of markup over the perceived marginal cost $c_i/(1-\tau)$, so that its retail price effectively reflects two layers of markup over the original marginal cost c_i.

In the hybrid mode, 3P sellers $i = 2, \ldots, n$ solve the same profit-maximization problem as in the pure marketplace mode. The platform chooses p_P to maximize the sum of its retail profit and commission revenue from 3P sales:

$$\max_{p_P} (p_P - c_P)D_P(\mathbf{p}) + \tau \sum_{i=2}^{n} p_i D_i(\mathbf{p}).$$

The first-order condition is

$$p_P - \left(c_P + \tau \sum_{i=2}^{n} p_i \delta_{Pi}(\mathbf{p})\right) = -\frac{D_P(\mathbf{p})}{\partial D_P(\mathbf{p})/\partial p_P}, \tag{2}$$

where

$$\delta_{ij} \equiv -\frac{\partial D_j(\mathbf{p})/\partial p_i}{\partial D_i(\mathbf{p})/\partial p_i}, \qquad i, j \in \{1, \ldots, n\}, \quad i \neq j,$$

is the diversion ratio from product i to product j. It measures the fraction of the marginal demand lost by product i following an increase in its price that is captured by product j. Therefore, the additional term $\tau \sum_{i=2}^{n} p_i D_i(\mathbf{p})$ in the platform's profit function captures the second channel through which τ affects price competition, which we call the *revenue-sharing channel*. As Equation (2) shows, when the platform raises its own price, the platform internalizes the resulting diversion toward 3P products and the consequent increase in commission revenue.

Taken together, these two channels create an inherent asymmetry between the pricing incentives of the platform and those of 3P sellers. The existence of these two distinct pricing channels arise from the platform's dual role as both retailer and intermediary under an ad valorem commission structure, rather than from a particular functional form for demand.

3.3 Pricing Asymmetries

We now analyze how the interplay between the two pricing forces shapes price competition along two dimensions: equilibrium price levels and retailers' price adjustments in response to cost shocks.

3.3.1 Equilibrium Price Comparisons

Comparing the first-order conditions of 3P sellers and the platform in Equations (1) and (2), we see that the two channels generate opposing effects on the platform's retail price relative to 3P sellers' prices. The revenue-sharing channel creates upward pressure on the platform's price, as a higher own-product price diverts demand toward 3P products and thereby increases commission revenue. In contrast, because the platform's retail and intermediation activities are vertically integrated, it avoids the double-markup distortion faced by 3P sellers, creating downward pressure on its price relative to theirs. The equilibrium price ranking therefore depends on the relative strength of these two forces.

For analytical tractability, we henceforth assume that all retailers have the same marginal cost, $c_i = c, \forall i \in \{1, \ldots, n\}$, regardless of whether retailer i is the platform or a 3P seller. We further impose the following linear specification on the symmetric demand system:

$$D_i(\mathbf{p}) = \frac{1-\gamma-(1+(n-2)\gamma)\,p_i+\gamma\sum_{j\neq i}p_j}{(1-\gamma)\,(1+(n-1)\gamma)}, \qquad \forall i\in\{1,\ldots,n\},$$

where $\gamma \in (0,1)$ captures the degree of substitutability among products. A higher γ corresponds to closer substitutes.

Under the pure marketplace mode, we focus on the symmetric equilibrium, characterized by Equation (1) and denote the common equilibrium price with n competing retailers by $p_M^*(n)$. Under the hybrid mode, we consider the semi-symmetric equilibrium in which all 3P sellers charge the same price. This equilibrium is characterized by Equation (1) for $i = 2,\ldots,n$ and Equation (2) for the platform. We denote the platform's equilibrium price by $\hat{p}_P^*(n)$ and the common equilibrium price of 3P sellers by $\hat{p}_S^*(n)$. Throughout the analysis, parameters are restricted to configurations under which the relevant equilibria in both modes are interior, so that all sellers are active in equilibrium.

Proposition 1. *When $0 < \tau < \max\left\{0, 1-\frac{2c(1+(n-2)\gamma)}{(n-1)\gamma}\right\}$, the upward force from the revenue-sharing channel dominates the downward force from avoiding the double-markup distortion. Consequently:*

(i) the platform charges a higher equilibrium price than 3P sellers: $\hat{p}_P^*(n) > \hat{p}_S^*(n)$;

(ii) the platform's retail entry exerts less competitive pressure on 3P sellers' prices than entry by an otherwise identical 3P seller: $\hat{p}_S^*(n) - p_M^*(n-1) > p_M^*(n) - p_M^*(n-1)$.

For commission rates above this threshold, the downward force from avoiding the double-markup distortion dominates, and both inequalities are reversed.

This result may appear counterintuitive: the revenue-sharing effect dominates when the commission rate τ is relatively low, even though a lower commission rate mechanically reduces the platform's commission revenue per dollar of 3P sales. The intuition is that 3P sellers' perceived marginal cost, $\frac{c}{1-\tau}$, increases convexly with the commission τ. Consequently, as the commission rate rises, the downward pricing force associated with eliminating the double-markup distortion becomes increasingly strong. Only when the commission rate is sufficiently high does this force become strong enough to outweigh the revenue-sharing effect, leading the platform to charge a lower price than 3P sellers. The same condition under which the revenue-sharing effect dominates and the platform charges a higher price than 3P sellers under the hybrid mode also implies that platform entry exerts less competitive pressure on incumbent 3P prices than entry by an otherwise identical 3P seller.

In sum, the efficiency advantage from eliminating the double-markup distortion does not unambiguously lead the platform to price more aggressively. Which of these opposing forces dominates depends on the underlying market conditions. Greater product substitutability, a larger number of competitors, and a lower marginal cost all expand the range of commission rates over which the revenue-sharing effect dominates. Appendix A.2 provides numerical illustrations of how the relative importance of the two pricing forces varies across parameter configurations.

Proposition 2. *The platform's entry as a retailer strictly lowers the equilibrium prices charged by 3P sellers,* $\hat{p}_S^*(n) < p_M^*(n-1)$.

In our framework of price competition with differentiated products, the effect of platform entry on 3P sellers' prices can be decomposed into two components:

$$\underbrace{\hat{p}_S^*(n) - p_M^*(n-1)}_{\text{total effect of platform entry}} = \underbrace{p_M^*(n) - p_M^*(n-1)}_{\text{standard 3P entry effect}} + \underbrace{\hat{p}_S^*(n) - p_M^*(n)}_{\text{platform-specific pricing effect}}.$$

The first term on the right-hand side captures the standard entry effect associated with an increase in the number of competitors. The second term isolates the platform-specific pricing effect by holding the number of competitors fixed and comparing 3P sellers' equilibrium prices across the two modes. This difference is therefore driven by the platform's two distinctive pricing incentives relative to those of an otherwise identical 3P seller. Under the linear demand system, the standard entry effect is sufficiently negative to offset any upward platform-specific pricing effect. Consequently, the platform's retail entry strictly lowers the equilibrium prices charged by incumbent 3P sellers.

To conclude the analysis on equilibrium prices, we compare our results with those of Hagiu et al. (2022) and Anderson and Bedre Defolie (2024). Hagiu et al. (2022) consider asymmetric Bertrand competition among the platform, a superior seller, and several fringe sellers. In their model, the platform's entry as a retailer "price-squeezes" the superior seller, forcing it to charge a lower price, because the platform does not incur the fixed transaction fee paid by 3P sellers and may also have a product-quality advantage over the fringe sellers. Similarly, Proposition 2 shows that the platform's retail entry also always lowers 3P sellers' equilibrium prices in our model. However, this result does not by itself imply that the platform's own product "price-squeezes" 3P sellers. Such an outcome arises only when the platform's advantage from avoiding the double-markup distortion dominates the revenue-sharing effect, so that the platform-specific pricing effect is also negative. In this case, the platform charges a lower equilibrium price than 3P sellers and exerts additional downward pressure on their prices relative to entry by an otherwise identical 3P seller. Our analysis also identifies cases in which the revenue-sharing effect dominates, so that the platform does not exert such a price squeeze and instead charges a higher price than 3P sellers in the hybrid equilibrium. This possibility does not arise in Hagiu et al. (2022). Our analysis also differs from Anderson and Bedre Defolie (2024). Their monopolistic-competition framework shuts down the effect of entry on incumbent sellers' market power. Consequently, the effect of the platform's entry on 3P sellers' prices operates entirely through the platform's adjustment of the commission rate following its switch from the pure marketplace mode to the hybrid mode. Their framework therefore abstracts from the pricing interactions between the platform and 3P sellers that are central to our analysis.

3.3.2 Asymmetric Cost Pass-Through

We now investigate how the two pricing forces identified above generate asymmetries in the pass-through of cost shocks to the platform's and 3P sellers' prices under the hybrid mode. We allow retailers to have heterogeneous marginal costs c_i, while maintaining the linear demand specification. Under this specification, the diversion ratio is identical for every pair of distinct products, so we write $\delta \equiv \delta_{ij}$ for all $i, j \in \{1, \ldots, n\}$, $i \neq j$.

We first define the pass-through matrix for seller-specific cost shocks. Let $\hat{\mathbf{p}}^* = (\hat{p}_1^*, \ldots, \hat{p}_n^*)^\top$ denote the vector of equilibrium prices under the hybrid mode, and let $\mathbf{c} = (c_1, \ldots, c_n)^\top$ denote the vector of marginal costs. Then, $\rho \equiv \frac{\partial \hat{\mathbf{p}}^*}{\partial \mathbf{c}^\top}$ is the pass-through matrix, where each element $\rho_{ij} \equiv \frac{\partial \hat{p}_i^*}{\partial c_j}$ measures the pass-through of a shock to retailer j's marginal cost into retailer i's price. We refer to the diagonal elements ρ_{ii} as own-cost pass-through rates and the off-diagonal elements ρ_{ij}, $i \neq j$, as cross-cost pass-through rates.

To distinguish the two channels through which τ affects pass-through, we write τ as τ_D in the 3P sellers' first-order condition in Equation (1) to capture the double-markup channel, and as τ_R in the platform's first-order condition in Equation (2) to capture the revenue-sharing channel. Substituting the linear demand system into these equilibrium conditions yields the following result.

Proposition 3. Pass-Through Matrix. *The pass-through matrix under the hybrid mode is given by*

$$\rho \equiv \frac{\partial \hat{\mathbf{p}}^*}{\partial \mathbf{c}^\top} = M(\tau_R)^{-1} C(\tau_D),$$

where

$$M(\tau_R) \equiv \begin{bmatrix} 2 & -\delta(1+\tau_R)\mathbf{1}_{n-1}^\top \\ -\delta \mathbf{1}_{n-1} & (2+\delta) I_{n-1} - \delta J_{n-1} \end{bmatrix}, \; C(\tau_D) \equiv \begin{bmatrix} 1 & \mathbf{0}_{n-1}^\top \\ \mathbf{0}_{n-1} & \dfrac{1}{1-\tau_D} I_{n-1} \end{bmatrix}, \text{ and } \tau_R = \tau_D = \tau.\text{[11]}$$

This decomposition highlights the distinct roles that the two channels play in determining pass-through. First, through the double-markup channel, τ_D directly amplifies the cost shocks to 3P sellers by a factor of $1/(1-\tau_D)$, as captured by the lower-right block of $C(\tau_D)$. In contrast, the platform's cost shock is not amplified. Second, through the revenue-sharing channel, τ_R strengthens the platform's strategic response to 3P sellers' prices, thereby making its price more responsive to cost shocks affecting 3P sellers. This can be seen from the matrix $M(\tau_R)$, which governs the strategic interactions among retailers' prices. Letting $BR_i(\mathbf{p}_{-i})$ denote retailer i's best-response function implied by its first-order condition, we have $\frac{\partial BR_P}{\partial p_j} = \frac{\delta(1+\tau_R)}{2} > \frac{\delta}{2} = \frac{\partial BR_i}{\partial p_j}$, $i, j \in \{2, \dots, n\}$, $i \neq j$. The inequality arises because an increase in 3P seller j's price affects the platform's pricing incentive not only through standard strategic complementarity but also by increasing the opportunity cost $\tau\delta \sum_{i=2}^n p_i$ of selling its own product. Through the revenue-sharing channel, the platform therefore has an additional incentive to raise its own price in response to an increase in p_j. Taken together, the two channels determine equilibrium pass-through through the decomposition $\rho = M(\tau_R)^{-1} C(\tau_D)$: $C(\tau_D)$ determines the effective cost shocks, while $M(\tau_R)^{-1}$ maps these shocks into equilibrium price responses by accounting for strategic interactions among prices.

We next compare pass-through between the platform and 3P sellers for both seller-specific cost shocks and an industry-wide cost shock. To define the latter, let ε denote a common shock that raises the marginal costs of all retailers one-for-one, so that $\frac{d\mathbf{c}}{d\varepsilon} = \mathbf{1}_n$. The pass-through of ε to retailer i's equilibrium price is then given by

$$\rho_i \equiv \frac{d\hat{p}_i^*}{d\varepsilon} = \sum_{j=1}^n \frac{\partial \hat{p}_i^*}{\partial c_j} \frac{dc_j}{d\varepsilon} = \sum_{j=1}^n \rho_{ij}, \qquad \forall i \in \{1, \dots, n\}.$$

[11] Here, $\mathbf{1}_k$ and $\mathbf{0}_k$ denote the k-dimensional column vectors of ones and zeros, respectively, I_k denotes the $k \times k$ identity matrix, and $J_k \equiv \mathbf{1}_k \mathbf{1}_k^\top$ denotes the $k \times k$ matrix of ones.

Proposition 4. Pass-through Asymmetries. *Under $\tau_R = \tau_D = \tau \in (0,1)$, the pass-through rates exhibit the following asymmetries:*

(1) The platform's own-cost pass-through rate is lower than that of any 3P seller: $\rho_{PP} < \rho_{SS}$.

(2) If $n \geq 3$, then, for any two distinct 3P sellers $S, S' \in \{2, \ldots, n\}$, the cross-cost pass-through rates satisfy $\rho_{SP} < \rho_{SS'} < \rho_{PS}$: a 3P seller responds more strongly to another 3P seller's cost shock than to the platform's cost shock, whereas the platform responds more strongly to a 3P seller's cost shock than does another 3P seller.

(3) The pass-through rate of an industry-wide cost shock to the platform's price is lower than that to any 3P seller's price: $\rho_P < \rho_S$.

The intuition is simple. Before accounting for strategic feedback, a one-unit increase in the platform's marginal cost raises its price by $\frac{1}{2}$, whereas the same increase in a 3P seller's marginal cost raises the seller's price by $\frac{1}{2(1-\tau_D)} > \frac{1}{2}$. Thus, the double-markup channel associated with τ_D tends to make a 3P seller's own-cost pass-through exceed the platform's. By contrast, the revenue-sharing channel tends to increase the platform's own-cost pass-through because it strengthens the platform's strategic response to 3P sellers' prices. When both channels are governed by the same commission rate, the cost-shock amplification effect dominates because it operates directly on the 3P seller's pricing condition, whereas the revenue-sharing effect operates only indirectly through strategic price feedback, with the effect of the original cost shock attenuated at each stage of the feedback process. Hence, $\rho_{PP} < \rho_{SS}$.

Regarding the cross-cost pass-through asymmetries, consider first the responses of the platform and an unshocked 3P seller to the same 3P seller's cost shock. Both responses are subject to the same cost-shock amplification effect, but the platform responds more strongly because revenue sharing makes its best response more sensitive to the shocked seller's price. Hence, $\rho_{PS} > \rho_{SS'}$. Next, a 3P seller responds more strongly to another 3P seller's cost shock than to the platform's cost shock because the former is amplified by $\frac{1}{1-\tau_D}$ before entering the strategic price-feedback process, whereas the latter enters this process without such amplification. Hence, $\rho_{SS'} > \rho_{SP}$.

Finally, the seller-specific pass-through results allow us to interpret the common-shock pass-through asymmetry. An industry-wide cost shock combines each retailer's own-cost response with its responses to all rivals' cost shocks. The higher own-cost pass-through of 3P sellers tends to make their prices respond more strongly to the common shock than the platform's price does, whereas the cross-cost asymmetry works in the opposite direction. The key distinction is that an own-cost shock enters a retailer's pricing decision directly, whereas a rival's cost shock must first affect the rival's price and then pass through the retailer's strategic response to that price. The strategic responses to all rival prices, $\sum_{j \neq i} \partial BR_i / \partial p_j$, sum to $(n-1)\delta/2 < 1$ for any 3P seller and to $(n-1)\delta(1+\tau_R)/2 < 1$ for the platform. Thus, even in aggregate, rivals' cost shocks are attenuated by this additional transmission layer. Consequently, when the cost-amplification and revenue-sharing channels are governed by the same commission rate, the own-cost pass-through asymmetry dominates, implying $\rho_P < \rho_S$.

3.4 Empirical Implications

The theoretical analysis yields two key findings that guide our empirical analysis:

- **Finding 1 (Differential Competitive Effects of Entry):** Although platform entry intensifies competition relative to no entry, the platform does not necessarily compete as aggressively as an otherwise identical 3P entrant.

- **Finding 2 (Asymmetric Pass-Through):** The platform's retail price is less responsive to industry-wide cost shocks than 3P sellers' prices.

In the next sections, we examine whether these patterns are present in the Amazon marketplace data. To study the differential competitive effects of entry, in Section 4, we exploit repeated within-ASIN changes in Amazon's and 3P sellers' retail availability and estimate event-time patterns in 3P prices around these changes. Then, to examine the asymmetric pass-through implication, in Section 5, we use shocks to commodity prices to compare the responsiveness of Amazon's and 3P sellers' prices to exogenous input cost changes.

4 Empirical Evidence: Price Responses to Entry

Our objective in this section is to examine whether changes in 3P prices around Amazon entry differ systematically from those around 3P entry, a comparison motivated by Finding 1.

4.1 Empirical Approach

Estimating the price response to seller entry confronts a classic identification challenge: seller entry and exit are endogenous (Bresnahan and Reiss, 1991). In our setting, this identification problem would be particularly acute if we were to compare ASINs with and without an observed Amazon retail offer, because Amazon's retail presence is selective across products (see Fact 1 in Section 2.3) and likely correlated with product-specific demand, inventory, and 3P competitive conditions. Similar concerns apply to the entry and exit of 3P sellers.

To mitigate this concern, we instead exploit within-ASIN variation in seller availability over time, as documented in Fact 2. These within-ASIN changes are frequent, short-lived, and reversible: a given seller type may appear or disappear on the same ASIN multiple times, potentially reflecting stockouts or replenishment frictions. We therefore use "entry" and "exit" as shorthand for changes in observed seller availability, rather than permanent product-market entry or exit.

For each seller-availability transition, we focus on the response of the 3P price for the same ASIN. Because offers within an ASIN correspond to the same underlying product, they are naturally close substitutes. The resulting same-ASIN price response can therefore serve as a natural empirical counterpart to the 3P price response to entry highlighted by our model, allowing us to sidestep the need to define the broader set of substitute products. Product subcategories, each comprising multiple ASINs, are used only to organize the sample, rather than to delineate the set of competitors represented in our model.

Taken together, this setting naturally lends itself to an event-study approach: we examine short-window price dynamics around well-defined transitions in seller availability. We next describe this event-study design in detail.

Event definition. We define a clean "Amazon entry" event as a sequence in which (i) no Amazon offer is observed and the relevant 3P price is observed for at least K consecutive days before the event, and (ii) both an Amazon offer and the relevant 3P price are observed for at least K consecutive days afterward. We define "Amazon exit" analogously. Because we do not observe the identities of all active 3P sellers in the data, we define "3P entry" as one-unit changes in the 3P seller count. For 3P events, Amazon remains absent and the relevant 3P price remains observed throughout the pre- and post-event periods. This implies that the window length K defines our "qualifying" entry and exit events. In the baseline specification, $K = 10$, while alternative specifications are shown in the

Appendix.

Econometric specification. We perform two types of "sharp" event studies (Kleven, Landais and Søgaard, 2019) around symmetric event windows. The first focuses on Amazon entry and exit events to characterize how the lowest observed 3P price evolves around changes in Amazon's retail availability. The second stacks Amazon and 3P entry or exit events in the same regression, allowing us to directly compare the corresponding price responses. For each event e, let T_e denote the event date. Define relative event time as $r = t - T_e$.

For Amazon entry and exit events, we estimate the following event-time specification:

$$y^m_{er} = \sum_{\substack{j=-10 \\ j \neq -1}}^{10} \beta^m_j \mathbf{1}\{r = j\} + \alpha_e + \varepsilon^m_{er} \tag{3}$$

In our baseline analysis, the outcome variable y^m_{er} is the logarithm of the lowest observed 3P price under fulfillment mode $m \in \{\text{FBA}, \text{FBM}\}$. α_e is an event fixed effect[12], and the coefficient β^m_j measures the change in the log of the lowest 3P price at event day j, relative to day -1, within the same event. Our analysis focuses on the lowest observed 3P offer because Keepa does not report the full vector of 3P prices. Among the observed 3P offers, the lowest price provides a natural empirical proxy for the single equilibrium 3P price in our model. For brevity, we refer to $p^{3P,m}_{it}$ as the 3P price throughout the remainder of the analysis.

For the second specification, we stack Amazon and 3P events in a common regression, estimated separately for entry and exit. To rule out simultaneous changes in seller composition, we restrict the stacked sample to event windows in which the total number of active sellers (including Amazon when present) is constant within the pre- and post-event periods and changes by exactly one at the event date. Let $g_e \in \{A, 3P\}$ index the event type of event e, where A denotes an Amazon event and $3P$ a 3P seller-count event. Let $\mathbf{1}_{\{g_e = A\}}$ and $\mathbf{1}_{\{g_e = 3P\}}$ denote the corresponding event-type indicators. We estimate the following specification separately for entry and exit:

$$y^m_{er} = \sum_{\substack{j=-10 \\ j \neq -1}}^{10} \left[\beta^m_{j,A} \mathbf{1}_{\{r=j\}} \mathbf{1}_{\{g_e = A\}} + \beta^m_{j,3P} \mathbf{1}_{\{r=j\}} \mathbf{1}_{\{g_e = 3P\}} \right] + \alpha_e + \varepsilon^m_{er} \tag{4}$$

The primary outcome variable y^m_{er} is once again the logarithm of $p^{3P,m}_{it}$ for $m \in \{\text{FBA}, \text{FBM}\}$. The

[12] This fixed effect is more granular than ASIN fixed effects would be, given that ASINs may undergo multiple relevant entry or exit events. Using calendar-date and ASIN fixed effects instead yields very similar results.

coefficients $\beta^m_{j,A}$ and $\beta^m_{j,3P}$ trace the Amazon and 3P event-time paths, respectively. The formal Amazon-vs.-3P difference at event time j is $\beta^m_{j,A} - \beta^m_{j,3P}$.[13]

Sample Construction. We implement the main analysis in the coffee category, where Amazon retail participation is frequent and where most products have reviews, indicating high turnover (see Table 1).[14] For all analyses in this section, we restrict attention to ASINs that exhibit meaningful variation in seller composition (specifically, we require that, at any point in our sample, Amazon and 3P sellers are jointly active for at least 10 consecutive days, and only 3P sellers are active for at least 10 consecutive days). We moreover exclude the brand that displays a brand-level assortment change in our period of observation (as discussed in Section 2.3).

Identification and Diagnostics. A causal interpretation requires that, among qualifying events, the transition date is unrelated to contemporaneous or anticipated shocks to the counterfactual 3P prices, that 3P sellers do not adjust their behavior in anticipation, and that, absent the transition, 3P prices would remain at their pre-event paths over the short event window.

As shown in Fact 2 (Figure 2, see also Appendix C.2), Amazon's recurrent presence and absence spells have heterogeneous durations, which reduce concerns that permanent assortment changes or a few platform-wide shocks drive the variation. We further assess whether qualifying changes in Amazon's availability are predictable from observable pre-event price and demand dynamics using hazard models for the probability that Amazon enters or exits an ASIN on the following day, conditional on the product's current seller regime. Our estimates show little relation between Amazon transitions and recent changes in prices or sales rank, although the hazards vary with some lagged levels, weekdays, and marketplace activity (Tables C.3 and C.4). Finally, the pre-event coefficients reported below provide complementary diagnostics for anticipation and pre-existing trends and, in the stacked design, for differential pre-trends.

4.2 Effect of Amazon Entry and Exit on 3P Prices

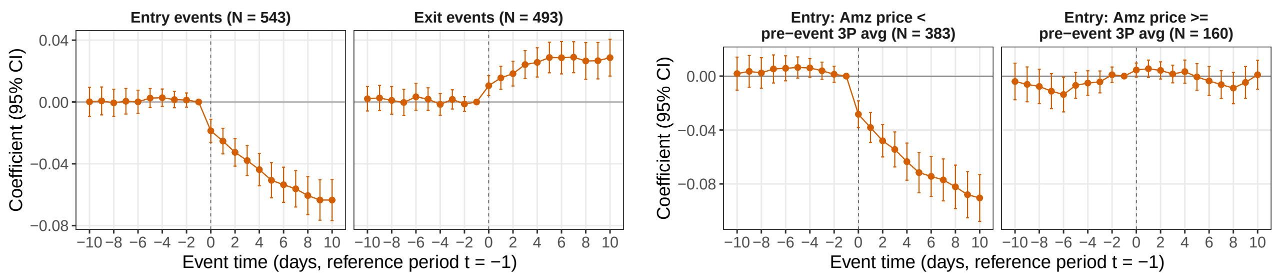
We first estimate event-study specifications for Amazon entry and exit (Equation (3)). Our baseline specification uses the lowest 3P price under FBA as an outcome variable, as sellers operating under

[13] As noted in Appendix B, prior to 2026, Keepa updated FBA and FBM prices at somewhat lower frequency than other price types. Event-time dynamics therefore cannot speak to the day-by-day speed of adjustment, but note that Musolff (2026) finds high re-pricing frequencies, suggesting that price adjustments may occur faster than our figures suggest.

[14] Unlike apparel or electronics, grocery categories like coffee are also not exposed to contemporaneous entry by low-cost import platforms such as Temu or Shein.

FBA are arguably closer substitutes to Amazon. Panel (a) in Figure 5 plots the estimated event-study coefficients, while the corresponding coefficient estimates and fit statistics are reported in Table D.3. The pre-event coefficients are small and jointly statistically indistinguishable from zero across the entry and exit specifications (Table D.1). For entry events, we document a strong decline in the lowest 3P price by almost 6% (3% for FBM prices). For exit events, we find the opposite pattern: mostly flat pre-trends are followed by an increase in the price by up to 3% (similar for FBM prices).

Figure 5: Event study of entry and exit by Amazon as a seller of a given ASIN; outcome variable: $\log\left(p^{3P,FBA}\right)$



(a) Lowest 3P price under FBA over event window.

(b) Split by Amazon's price at entry.

Notes: Figures show coefficient estimates stemming from Equation (3), as well as 95% confidence intervals computed using two-way clustered standard errors (ASIN × calendar-date). Panel (a) displays specifications (1) and (3) of Table D.3. Panel (b) shows a sample split of specification (1) using, first, observations in which Amazon enters with a price that is lower than the pre-event average lowest 3P FBA price, and second using observations in which Amazon enters with a price that is as least as high. All regressions are estimated using "qualifying" events in the daily coffee data, as defined earlier.

One possibility is that the 3P price response to Amazon entry reflects Amazon's platform identity rather than the competitive position of its offer. To assess whether the response varies with Amazon's entry price, Panel (b) splits entry events into two groups: Amazon enters with a relatively low price (defined as a price below the pre-event average of 3P FBA prices; 383 out of 543 events) or with a price that is at or above (160 out of 543 events). The estimates show that the responses are strikingly different: price declines upon entry are significant only if Amazon enters with a price that is lower than the average lowest FBA price in the pre-window period.[15] The substantially stronger price response when Amazon enters at a relatively low price is consistent with 3P sellers responding to the competitive position of Amazon's offer and, more generally, with the price-mediated competitive channel illustrated by the theoretical model.[16]

[15] The analogous finding arises in the case of exit events (see Figure D.3 in the Appendix).

[16] Panel (a) in Figure D.4 shows that analysis using lowest 3P FBM prices exhibits qualitatively a pattern similar to the FBA estimates. Figure D.3 displays further subsample analyses by showing the analogue of Figure 5 (b) for exit, as well as an entry analysis split by seller count.

We supplement these results with two additional checks. First, on the event date, total seller count rises by about one at Amazon entry and falls by about one at exit, consistent with no additional same-day change in 3P participation alongside Amazon entry or exit (see panel (b) of Figure D.4). Second, demand shocks are potential confounders that could cause price changes. To gauge this possibility, we examine the patterns in sales ranks, which are informative about demand dynamics around events. A lower sales rank indicates stronger sales performance. The results show that sales rank stays roughly flat during the pre-period (see panel (c) of Figure D.4 in the Appendix), providing little evidence of systematic pre-existing demand trends around the qualifying transitions.

To summarize, our findings show that intermittent changes in Amazon's retail availability are associated with sizable and reversible movements in the lowest 3P price, with stronger responses when Amazon enters at a relatively competitive price. This heterogeneity is consistent with a price-mediated competitive channel.

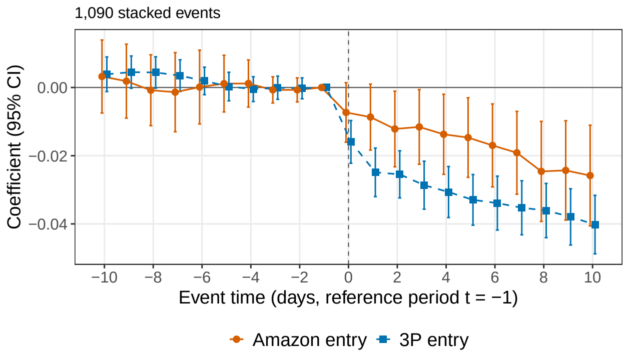
4.3 Differential Effects of Amazon and 3P Entry and Exit

Our objective is to compare changes in the lowest observed 3P price around Amazon entry (exit) with those around 3P entry (exit) by estimating equation 4.

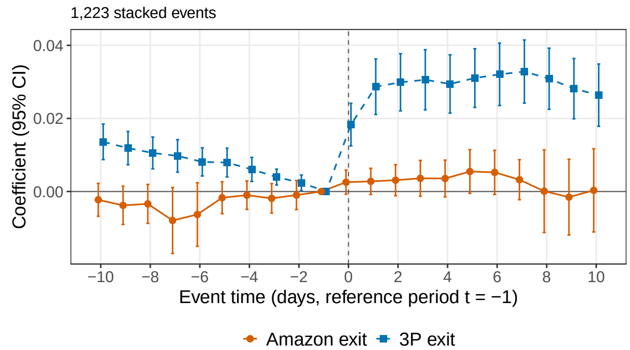
Figure 6 plots the estimates, while Table 2 reports average post-event effects and contrasts the coefficients of different event types. Across event types, the point estimates suggest smaller 3P price movements after Amazon entry/exit than after 3P seller entry/exit. The contrast is strongest when considering prices under FBA: over days 0–10, the lowest 3P price falls by around 3.1% after 3P entry and 1.6% after Amazon entry. It rises by nearly 3% after 3P exit, but stays essentially unchanged after Amazon exit. Given the apparent pre-trends for the FBA exit specification, we prefer the FBM estimates which point in the same direction while having flat pre-event paths; however, here, the Amazon–3P differences are not statistically significant. Overall, we read the stacked evidence as consistent with weaker Amazon-induced price responses. The patterns therefore provide suggestive evidence for the theoretical possibility that platform entry may exert less downward pressure on 3P prices than 3P entry.

One data limitation is that we do not observe the identities of all active 3P sellers. Therefore, around 3P entry and exit, changes in the lowest observed 3P price may reflect both repricing by continuing sellers and changes in the set of offers over which the minimum is taken. The stacked

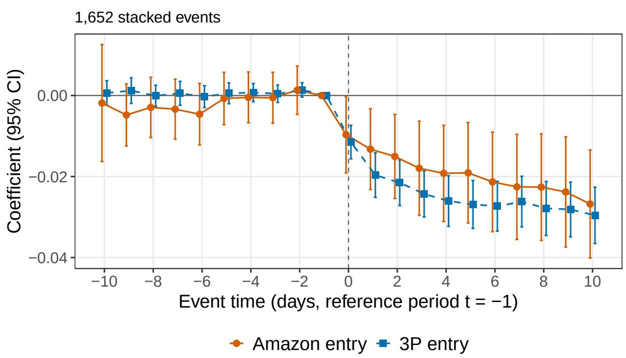
Figure 6: Stacked event study of entry and exit by Amazon and a 3P seller; outcome variables: $\log(p^{3P,FBA})$ and $\log(p^{3P,FBM})$



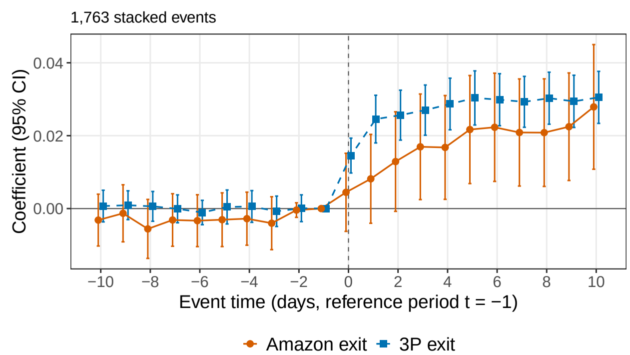
(a) Entry, FBA.



(b) Exit, FBA.



(c) Entry, FBM.



(d) Exit, FBM.

Notes: The figure shows coefficient estimates stemming from Equation (4) with the lowest-priced 3P offer as an outcome variable, as well as 95% confidence intervals computed using standard errors two-way clustered by ASIN and calendar date. All regressions are estimated using "qualifying" events in the daily coffee data, as defined in the text.

Table 2: Average post-event effects in stacked Amazon-vs-3P event studies

Event	3P price	Window	3P event	Amzn event	|Amzn| − |3P|	p-value: |Amzn| = |3P|	Events: Amzn / 3P
Entry	FBA	0–10	-3.11*** (0.35)	-1.63*** (0.56)	-1.48** (0.65)	0.023	147 / 943
Entry	FBA	3–10	-3.45*** (0.38)	-1.89*** (0.61)	-1.56** (0.71)	0.028	147 / 943
Entry	FBM	0–10	-2.45*** (0.29)	-1.92*** (0.57)	-0.52 (0.61)	0.387	207 / 1445
Entry	FBM	3–10	-2.70*** (0.31)	-2.17*** (0.62)	-0.54 (0.67)	0.420	207 / 1445
Exit	FBA	0–10	2.89*** (0.38)	0.26 (0.26)	-2.63*** (0.46)	0.000	222 / 1001
Exit	FBA	3–10	3.02*** (0.41)	0.25 (0.32)	-2.77*** (0.51)	0.000	222 / 1001
Exit	FBM	0–10	2.73*** (0.33)	1.78*** (0.68)	-0.95 (0.76)	0.209	232 / 1531
Exit	FBM	3–10	2.94*** (0.35)	2.12*** (0.74)	-0.82 (0.82)	0.317	232 / 1531

Notes: Estimates are average event-time coefficients over the indicated post-event window, relative to $t = -1$. Estimates and standard errors are reported as log points multiplied by 100. 3P denotes the 3P-seller-event path; Amzn denotes the implied Amazon-event path. $|\text{Amzn}| - |\text{3P}|$ compares the absolute magnitudes of the implied Amazon-event and 3P-event paths; negative values indicate that the Amazon-event response is smaller in absolute value. Event counts are unique event IDs in the estimation sample. Clustered (ASIN $\times$ calendar date) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

estimates should thus not be interpreted as isolating incumbent repricing. We do not view this limitation as undermining the economic content of the comparison, however, as long as the lowest observed 3P price is interpreted as a reduced-form measure of the competitive 3P price frontier. Under this interpretation, both repricing by existing sellers and the price introduced or removed by a 3P entrant contribute to the overall effect of entry or exit on the competitive price available in the marketplace.

4.4 Robustness and Additional Results

Alternative qualifying-spell requirements. Our baseline stacked event studies set $K = 10$, requiring the relevant seller compositions to persist for at least 10 consecutive days around each qualifying transition. We assess the sensitivity of the results to this requirement by re-estimating the event studies using $K = 5$ and $K = 15$. Since K enters the qualifying-event definition itself, varying K also changes the set of transitions included in the estimation. Figure D.1 in the Appendix reports the corresponding estimates. The baseline patterns are generally more closely reproduced under the more restrictive definition $K = 15$ than under $K = 5$. For FBA prices, the $K = 15$ estimates preserve the baseline ordering for both entry and exit. The FBM entry estimates also remain relatively close across Amazon and 3P events, as in the baseline specification, although the FBM exit differential is less stable and is estimated imprecisely, which is consistent with the smaller number of events in this case. By contrast, when admitting shorter-lived regimes under $K = 5$, the FBA entry differential reverses sign, while the difference between Amazon and 3P entry under FBM becomes substantially larger than in the baseline. These results suggest that the estimated entry differential is sensitive to the less restrictive qualifying-spell requirement, whereas transitions between more persistent regimes produce patterns closer to the baseline estimates.

Additional market outcomes. Panels (c) and (d) of Figure D.4 examine changes in product-level sales rank and Buy Box assignment around Amazon entry and exit. As noted above, a lower sales rank indicates stronger sales performance; thus, the decline in log sales rank following Amazon entry, together with the corresponding increase following exit, is consistent with Amazon availability being associated with improved sales performance or visibility for the ASIN. However, these estimates do not reveal how sales are allocated between Amazon and 3P sellers or distinguish between demand expansion and substitution across offers. Amazon entry also raises the probability that

Amazon occupies the Buy Box by approximately 30 percentage points, while Amazon exit produces a corresponding decline. It shows that Amazon's participation may change which offer is prominently displayed to consumers, but Amazon does not automatically win the Buy Box whenever it enters.

Alternative specifications. For the effects pertaining to Amazon's presence, we assess whether a similar association holds in other product categories and outside our somewhat selective qualifying events. A two-way fixed effects specification (Tables D.4 and D.5) yields uniformly negative and statistically significant associations of Amazon presence with FBA prices, with magnitudes ranging from 2.8% for TVs to 10.8% for pasta. Hence, the broader pattern is consistent with our evidence on qualifying events in the coffee category.

5 Empirical Evidence: Asymmetric Pass-Through of Cost Shocks

Our objective in this section is to examine how Amazon and 3P prices respond to common cost shocks and whether their pass-through rates differ, thereby providing empirical evidence for Finding 2.

5.1 Empirical Approach

To identify the differential price responses, we exploit variation in upstream commodity costs that is plausibly exogenous to individual ASINs and provides a common upstream cost shifter for Amazon and 3P sellers.

We restrict our analysis to ground coffee and baking cocoa, a choice motivated by three considerations. First, their main raw inputs – coffee beans and cocoa beans – are major internationally traded commodities with transparent and frequently observed market prices. Second, these raw commodities are salient and economically important inputs into the corresponding downstream products in our data. Third, both commodity prices exhibit substantial variation over our sample period, providing meaningful time-series variation for estimating pass-through. Taken together, these features make coffee- and cocoa-bean prices well suited as common upstream cost shifters for comparing the pricing responses of Amazon and 3P sellers.[17]

[17] Prior work has also used coffee-bean price shocks to study pass-through (Bettendorf and Verboven, 2000; Nakamura and Zerom, 2010; Bonnet, Dubois, Villas-Boas and Klapper, 2013; Sangani, 2026). We do not study other products affected by cocoa-bean prices, such as chocolate bars, because these products often involve a large share of so-

We obtain daily commodity-price data from Trading Economics (`tradingeconomics.com`) and the International Cocoa Organization (`www.icco.org`; see Appendix B.4). To express commodity and retail prices in comparable units, we use conversion factors that account for roasting losses in coffee and bean-to-powder processing yields in cocoa, thereby translating raw-bean prices into implied commodity input costs per gram of the corresponding downstream product. This approach follows the related pass-through literature (e.g., Nakamura and Zerom, 2010; Sangani, 2026; Bonnet et al., 2013).[18] Panels (a) and (b) of Figure 7 plot the resulting commodity-cost series for our sample period of January 2021-March 2025. Importantly, the sharp increases in coffee and cocoa prices observed over our sample period were widely attributed to adverse supply conditions: poor weather in Brazil in the case of coffee, and poor weather and crop disease in Côte d'Ivoire and Ghana in the case of cocoa, with the latter set of disruptions referred to as the "cocoa crisis" by the financial press.[19] This supply-side origin supports the interpretation of the commodity price changes as common upstream cost shifters.

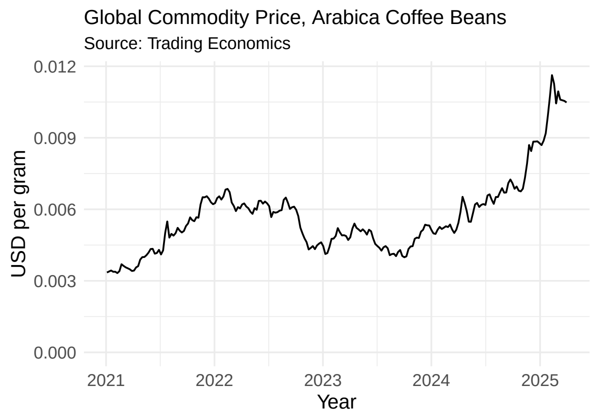
We next provide a descriptive summary of posted-price changes. Cross-sectional means or medians would combine within-ASIN repricing with changes in the mix of ASINs offered by Amazon and 3P sellers. Panels (c) and (d) of Figure 7 therefore report chained Jevons indices for Amazon's posted price and the lowest observed 3P offer, which are less sensitive to changes in product composition. For each price series and pair of adjacent calendar weeks, we compare an ASIN's price with its own price in the preceding week, conditional on that price series being observed for the ASIN in both weeks. We then take the equal-weighted geometric mean of these within-ASIN price relatives and chain the resulting weekly links. Both product categories exhibit an upward trend in within-ASIN posted prices against the backdrop of large increases in commodity prices. In coffee, the index for the lowest 3P offer rises more than the Amazon index. In cocoa, both indices rise substantially and finish at broadly similar levels, although their paths differ. Taken together, these descriptive patterns point to a certain degree of heterogeneity in both the magnitude and timing of price adjustment across seller types: the divergence is particularly visible in coffee, while in cocoa

called "local costs" (Nakamura and Zerom, 2010), such as advertising expenditures. In addition, chocolate manufacturers have reportedly adjusted product recipes in response to input-price shocks (see `https://www.ft.com/content/169c9fca-1a7a-4c0f-84e9-398777335629`, last accessed 26/02/2025), which further limits the suitability of these products for studying price pass-through.

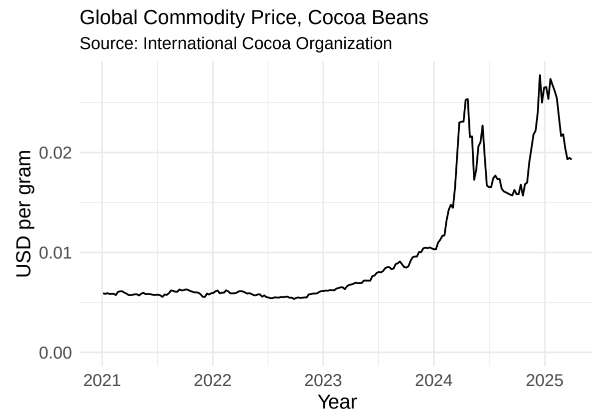
[18] The adopted conversion factors and their sources are documented in Appendix Table B.3.

[19] See `https://www.ft.com/content/a08997cf-8e19-41e5-af4f-ef1e3ebdb236?syn-25a6b1a6=1` and `https://www.ft.com/content/8a917bea-8ce3-11e3-ad57-00144feab7de` (last accessed October 10, 2025). We verified that no material changes in U.S. tariffs on the relevant coffee and cocoa inputs occurred during the sample period.

Figure 7: Commodity prices and price indices of products sold on Amazon

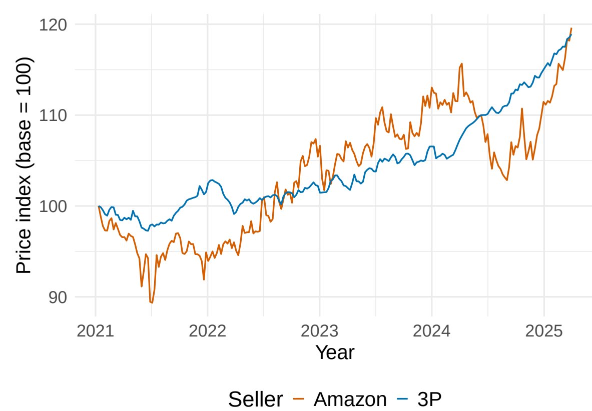


(a) Coffee.



(b) Cocoa.

(c) Coffee, chained price index.



(d) Cocoa, chained price index.

Notes: Panels (a) and (b) plot weekly average prices for Arabica coffee beans and cocoa beans, respectively, adjusted by the conversion factors discussed in Appendix B.4. Panels (c) and (d) report seller-type-specific chained matched-ASIN Jevons price indices. For seller type s and adjacent calendar weeks $t-1$ and t, let $\mathcal{M}_{s,t}$ denote the set of ASINs for which the relevant weekly average posted price is observed in both weeks, and let $N_{s,t} = |\mathcal{M}_{s,t}|$. The bilateral link is $J_{s,t} = \exp\left\{\frac{1}{N_{s,t}}\sum_{i\in\mathcal{M}_{s,t}}\left[\log(p_{i,s,t}) - \log(p_{i,s,t-1})\right]\right\}$. Thus, each ASIN for which the relevant price series is observed in both adjacent weeks receives equal weight in that weekly link. We chain the adjacent-week links and normalize each series to 100 in the first full sample week of 2021. An ASIN observed in only one of two adjacent weeks does not enter that bilateral link. The set $\mathcal{M}_{s,t}$ may change across links, so the indices are rolling adjacent-week indices rather than fixed-basket indices over the full sample period.

similar cumulative changes mask different adjustment paths. Because the seller-specific indices may rely on different matched-ASIN sets, we next turn to the regression analysis for formal evidence on differential pass-through.[20]

We treat coffee as our primary empirical setting, as its larger sample allows for more precise estimates, and as the coffee commodity price exhibits more variation across the whole sample period.

[20]A common observation is that rising prices can affect package sizes ("shrinkflation"); however, we do not find any evidence of a change in the prevailing package size distribution over the sample period.

Cocoa provides a second application based on a second, independent commodity-cost shock.

Econometric specification. We estimate pass-through using a standard distributed-lag specification, which is widely used in the pass-through literature (e.g., Alvarez, Cavallo, MacKay and Mengano, 2026; Sangani, 2026). This specification provides a flexible reduced-form characterization of the dynamic response of retail prices to commodity-cost shocks. In particular, as our goal is to compare pass-through across seller types for the same ASIN-weeks, we stack Amazon's posted price and the lowest 3P seller price into a seller-type panel. The stacked specification estimates both price responses on the same set of observed ASIN-week cells, thereby holding fixed the ASIN-week composition of the estimation sample across seller types.

Let $s \in \{3P, A\}$ index seller type, where A denotes Amazon, and let $\mathbf{1}_{\{s=A\}}$ indicate that the seller type is Amazon. In our baseline specification, we retain only ASIN-week cells for which the weekly price change is observed for *both* seller types – i.e., where they have common support over ASIN-weeks – so that each retained ASIN-week contributes one observation for Amazon and one for the 3P offer. For each product category and each window length $K \in \{6, 12, 24, 48\}$, we estimate

$$\Delta p_{ist} = \sum_{k=0}^{K-1} \beta_k \Delta c_{t-k} + \sum_{k=0}^{K-1} \delta_k \left(\mathbf{1}_{\{s=A\}} \times \Delta c_{t-k} \right) + \mu_{is} + \varepsilon_{ist}, \tag{5}$$

where Δp_{ist} denotes the change in the posted retail price per gram for ASIN i and seller type s between weeks $t = 1$ and t. For $s = 3P$, the price indicates is the lowest observed 3P price, while for $s = A$, the price corresponds to Amazon's posted price. The variable Δc_t is the weekly change in the relevant raw-commodity price in dollars per gram after using the abovementioned the conversion factors (Appendix B.4). The fixed effect μ_{is} is specific to an ASIN-seller-type price series and allows Amazon's and the 3P price series for the same ASIN to have different average weekly price changes. All pass-through specifications use the same 222-week retail-price window, with commodity-cost lags extending into 2020.

For 3P sellers, $\mathbf{1}_{\{s=A\}} = 0$, so β_k is the pass-through of the commodity-cost change at lag k. For

Amazon, the corresponding pass-through is $\beta_k + \delta_k$. The cumulative pass-through measures are:

$$\widehat{\rho}_{3P}(K) \equiv \sum_{k=0}^{K-1} \widehat{\beta}_k, \tag{6}$$

$$\widehat{\rho}_{A}(K) \equiv \sum_{k=0}^{K-1} \left(\widehat{\beta}_k + \widehat{\delta}_k\right), \tag{7}$$

$$\widehat{\rho}_{A}(K) - \widehat{\rho}_{3P}(K) = \sum_{k=0}^{K-1} \widehat{\delta}_k. \tag{8}$$

Thus, equation (8) characterizes the cumulative Amazon-minus-3P pass-through differential, and a negative value indicates that Amazon passes through less of the commodity-cost shock than 3P sellers. Given that downstream prices and upstream commodity costs are measured in the same units in our data, the specification has a direct economic interpretation: the sum of the coefficients measures the cumulative pass-through of a one-unit increase in commodity cost per gram into the product price per gram.

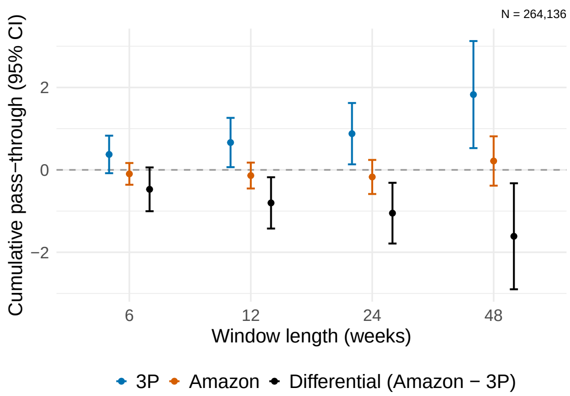
In light of Sangani (2026), who shows that for common commodity shocks, pass-through may appear incomplete in logs even when it is close to complete in levels, we take the levels specification as our baseline. We confirm that first differences in commodity prices are stationary for both cocoa and coffee using an augmented Dickey-Fuller test.

5.2 Differential Cost Pass-Through by Seller Type

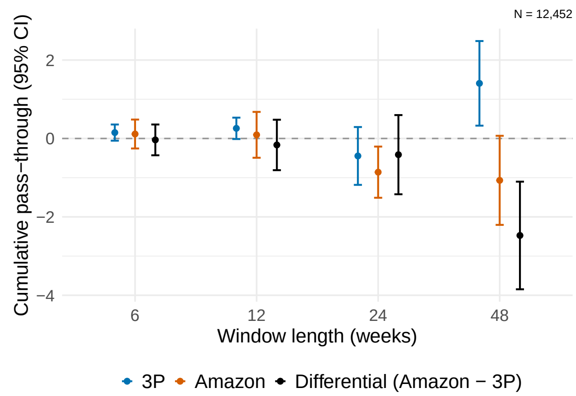
Figure 8 and Table 3 report the cumulative pass-through estimates from the baseline specification. The central result is a negative Amazon-minus-3P pass-through differential: conditional on the same set of ASIN-week observations, Amazon transmits less of the common commodity-cost shock into posted retail prices than the lowest 3P offer does. To the extent that these commodity-price changes generate comparable input-cost shocks across seller types, the observed differential is consistent with the asymmetric common-cost pass-through in Finding 2.

The evidence is strongest for coffee. The estimated Amazon-minus-3P differential is negative at every horizon and is statistically significant at the 12-, 24-, and 48-week horizons. The gap persists as the horizon lengthens, suggesting that it is not eliminated by allowing additional time for Amazon's price adjustment. At the 48-week horizon, the estimated cumulative pass-through is 1.827 for 3P sellers and 0.216 for Amazon, corresponding to an Amazon-minus-3P differential of -1.612. In line with Sangani (2026), long-run pass-through in levels is statistically indistinguishable from one for

Figure 8: Cumulative commodity-cost pass-through by seller type



(a) Coffee.



(b) Cocoa.

Notes: Cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include ASIN × seller-type fixed effects. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors clustered by ASIN and calendar week.

Table 3: Cumulative commodity-cost pass-through by seller type

	$K=6$	$K=12$	$K=24$	$K=48$
Panel A. Coffee				
3P	0.375	0.664**	0.879**	1.827***
	(0.232)	(0.305)	(0.380)	(0.663)
Amazon	-0.097	-0.138	-0.172	0.216
	(0.134)	(0.160)	(0.211)	(0.305)
Differential (Amazon − 3P)	-0.472*	-0.801**	-1.050***	-1.612**
	(0.271)	(0.318)	(0.376)	(0.656)
Observations	264,136	264,136	264,136	264,136
Panel B. Cocoa				
3P	0.150	0.258*	-0.446	1.406**
	(0.105)	(0.139)	(0.376)	(0.550)
Amazon	0.114	0.093	-0.859***	-1.068*
	(0.188)	(0.298)	(0.333)	(0.580)
Differential (Amazon − 3P)	-0.036	-0.165	-0.413	-2.474***
	(0.200)	(0.328)	(0.515)	(0.700)
Observations	12,452	12,452	12,452	12,452

Notes: Distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs; each column is a separate regression at window length K. 3P reports $\sum_{k=0}^{K-1}\widehat{\beta}_k$, Amazon $\sum_{k=0}^{K-1}(\widehat{\beta}_k+\widehat{\delta}_k)$, and the Differential $\sum_{k=0}^{K-1}\widehat{\delta}_k$; negative values indicate lower pass-through by Amazon. Prices and commodity costs are in dollars per gram (commodity prices converted at 1.235 for coffee and 2.44 for cocoa), so one denotes full pass-through. The sample is ASIN–week cells with both seller types' price changes observed, stacked so each cell contributes one Amazon and one 3P observation; all regressions include ASIN × seller-type fixed effects. Standard errors clustered by ASIN and calendar week in parentheses. $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.10$.

3P sellers, but not for Amazon.

The estimates for cocoa point in the same qualitative direction but are, given the much smaller sample size, considerably less precise; the Amazon-minus-3P differential is statistically distinguishable from zero only at the 48-week horizon.

5.3 Robustness and Alternative Outcome Variables

We assess the sensitivity of the baseline seller-type differential to using different subsamples, functional forms, and relaxing the common-support restriction. We then consider Buy Box-winning prices as an alternative, consumer-facing price measure.

Subsample analyses. The divergence between 3P and Amazon pass-through is not driven by particularly high-priced products; in fact, it tends to become stronger when excluding products that are "consistently" high-priced, defined as products whose median prices are strictly higher than the 80th percentile of all prices (Figure E.1). Given that 3P and Amazon presence within a given ASIN is often temporary, we also estimate our regressions on a subsample where we restrict to ASINs with at least 48 consecutive weeks of joint Amazon and 3P availability. The Amazon-minus-3P differential remains negative (Figure E.2): at the 48-week horizon it is nearly unchanged in both categories. Intermediate-horizon estimates become less precise for coffee, whereas the 3P vs. Amazon difference becomes more pronounced for cocoa. Finally, we conduct the estimations on the set of products with any user reviews or with a consistently lower sales rank, indicating higher turnover (Figures E.3 and E.4).

Alternative specifications. The pass-through difference remains substantial even when we relax the requirement that Amazon and 3P prices be jointly observed and instead estimate the pass-through regressions separately for Amazon and the lowest 3P prices (see Figure E.5). In the online appendix, we report the analogous log specification. The results are consistent with the findings of Sangani (2026): for 3P sellers, pass-through tends to be complete in levels, but less so in logs.

Buy Box-winning price. We also estimate seller-specific pass-through using Buy Box-winning prices. While this price arguably captures the price that most consumers will in the end pay when making a purchasing decision, the Buy Box price captures not only the pricing decision, but also the algorithm's ranking. Because only one offer can occupy the Buy Box at a time, the Amazon and 3P specifications necessarily are estimated on separate ASIN-weeks. For coffee, all estimates are statistically insignificant. For the case of cocoa, while estimates are noisier, we continue to obtain a significantly positive cumulative pass-through estimate at a 48-week horizon for the 3P Buy Box price, but its point estimate is substantially smaller than the corresponding baseline 3P estimate, while the estimate for Amazon is much lower. Across both categories, the attenuation of 3P pass-through when focusing on Buy Box-winning offers is broadly consistent with the price sensitivity of Buy Box assignment documented in prior work (Chen et al., 2016; Lee and Musolff, 2026). One possible interpretation is that 3P offers with larger price responses become less likely to remain represented among Buy Box winners as their prices rise, although these regressions do not directly identify changes in Buy Box ownership.

6 Conclusion

Our paper highlights the distinct pricing incentives that arise when a platform both retails on its own marketplace and earns commissions on 3P sales. In our model, this structure generates two key implications for pricing on the platform: platform entry can exert weaker competitive pressure on 3P prices than 3P entry, and common cost shocks generate larger price responses by 3P sellers than by the platform. Both implications reflect the interaction of hybrid operation and the ad valorem commission. Using Amazon U.S. data, we document reduced-form evidence consistent with both implications. 3P prices tend to fall less following Amazon entry than following 3P entry, and to rise less following Amazon exit than following 3P exit. In response to common commodity-cost shocks, 3P prices exhibit significantly greater pass-through than Amazon prices, with Amazon's estimated response statistically indistinguishable from zero or negative at different horizons.

More broadly, our results show that the competitive implications of hybrid operation extend beyond platform-controlled steering or prominence. Even absent exclusionary conduct, the platform's role as both retailer and commission-earning intermediary can alter the competitive environment faced by 3P sellers. In response to antitrust scrutiny of its hybrid mode, Amazon has defended this model on the ground that direct competition between its own retail offers and 3P sellers lowers prices and benefits consumers (Amazon.com Inc., 2023). Our findings suggest a more nuanced view of this argument: Amazon entry does intensify price competition relative to no entry, but the resulting pressure may be weaker than that generated by 3P entry. Moreover, our asymmetric pass-through result shows that hybrid organization can shape how common cost shocks are transmitted across seller types, causing platform and 3P sellers to bear and pass on such shocks differently. As online retail expands and platforms increasingly combine marketplace intermediation with their own retail activity, understanding how hybrid organization shapes cost pass-through may therefore have implications for inflation as well as for the pass-through of monetary policy.

Addressing these policy questions quantitatively requires a structural framework beyond the stylized model and reduced-form empirical design developed in this paper. A natural direction for future work is therefore to evaluate how alternative commission regimes or other platform regulations affect pass-through, price competition, and consumer welfare.

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Appendix

A Additional Theoretical Results and Proofs

A.1 Proofs of Results in Section 3

Proof of Proposition 1

Proof. Under the pure marketplace mode with n retailers, the symmetric equilibrium $\mathbf{p}_M^* = (p_M^*(n), \ldots, p_M^*(n))$ is characterized by

$$D_i(\mathbf{p}_M^*) + \left(p_M^*(n) - \frac{c}{1-\tau}\right)\frac{\partial D_i(\mathbf{p}_M^*)}{\partial p_i} = 0.$$

Substituting the linear demand function yields

$$p_M^*(n) = \frac{(1-\gamma)(1-\tau) + c\,(1+(n-2)\gamma)}{(1-\tau)\,(2+(n-3)\gamma)}.$$

Denote the semi-symmetric equilibrium under the hybrid mode by $\hat{\mathbf{p}}^* = (\hat{p}_P^*(n), \hat{p}_S^*(n), \ldots, \hat{p}_S^*(n))$. Let $D_P(\hat{\mathbf{p}}^*) = D_1(\hat{\mathbf{p}}^*)$ denote equilibrium demand for the platform's own product and let $D_S(\hat{\mathbf{p}}^*) = D_i(\hat{\mathbf{p}}^*)$ for all $i \in \{2, \ldots, n\}$ denote equilibrium demand for each 3P seller. The equilibrium is characterized by

$$\begin{cases} D_P(\hat{\mathbf{p}}^*) + (\hat{p}_P^*(n) - c)\,\dfrac{\partial D_P(\hat{\mathbf{p}}^*)}{\partial p_P} + \tau(n-1)\hat{p}_S^*(n)\dfrac{\partial D_S(\hat{\mathbf{p}}^*)}{\partial p_P} = 0, \\ D_S(\hat{\mathbf{p}}^*) + \left(\hat{p}_S^*(n) - \dfrac{c}{1-\tau}\right)\dfrac{\partial D_S(\hat{\mathbf{p}}^*)}{\partial p_S} = 0. \end{cases}$$

The solution is:

$$\begin{cases} \hat{p}_P^*(n) = \frac{c(1+(n-2)\gamma)[2(1-\tau)+\gamma(2n-3+\tau)]+(1-\gamma)(1-\tau)[2+\gamma(2n-3+(n-1)\tau)]}{(1-\tau)[4+6(n-2)\gamma+\gamma^2(2n^2-9n+9-(n-1)\tau)]}, \\ \hat{p}_S^*(n) = \frac{c(1+(n-2)\gamma)[2+\gamma(2n-3-\tau)]+(1-\gamma)(1-\tau)[2+(2n-3)\gamma]}{(1-\tau)[4+6(n-2)\gamma+\gamma^2(2n^2-9n+9-(n-1)\tau)]}. \end{cases}$$

Direct comparison yields

$$\hat{p}_P^*(n) \geq \hat{p}_S^*(n) \iff \hat{p}_S^*(n) \geq p_M^*(n) \iff \tau \leq 1 - \frac{2c\,(1+(n-2)\gamma)}{(n-1)\gamma}.$$

Finally, we explain why the two price comparisons share the same threshold. Evaluating the platform's first-order condition at the symmetric pure-marketplace price profile $\mathbf{p}_M^*(n)$ and setting the resulting marginal-profit equal to zero, we obtain

$$\begin{aligned} & D_P(\mathbf{p}_M^*) + (p_M^*(n) - c)\,\frac{\partial D_P(\mathbf{p}_M^*)}{\partial p_P} + \tau(n-1)p_M^*(n)\frac{\partial D_S(\mathbf{p}_M^*)}{\partial p_P} = 0 \\ \iff \quad & \tau = 1 - \frac{2c\,(1+(n-2)\gamma)}{(n-1)\gamma}. \end{aligned}$$

This implies that, when

$$\tau = 1 - \frac{2c\,(1+(n-2)\gamma)}{(n-1)\gamma},$$

the revenue-sharing effect exactly offsets the effect associated with eliminating the double-markup distortion. The platform therefore finds it optimal to charge the pure-marketplace equilibrium price, and

$$\hat{p}_P^*(n) = \hat{p}_S^*(n) = p_M^*(n).$$

When

$$\tau < 1 - \frac{2c\,(1+(n-2)\gamma)}{(n-1)\gamma},$$

the revenue-sharing effect dominates the effect associated with eliminating the double-markup distortion, so the platform finds it profitable to raise its price above $p_M^*(n)$. Because prices are strategic complements under the linear demand system, 3P sellers respond by raising their prices as well. Consequently,

$$\hat{p}_P^*(n) > \hat{p}_S^*(n) > p_M^*(n).$$

When

$$\tau > 1 - \frac{2c\,(1+(n-2)\gamma)}{(n-1)\gamma},$$

the effect associated with eliminating the double-markup distortion dominates, so the platform finds it profitable to set a price below $p_M^*(n)$. The 3P sellers respond by lowering their prices. Consequently, the hybrid-mode equilibrium features lower prices for all retailers than the pure-marketplace equilibrium:

$$\hat{p}_P^*(n) < \hat{p}_S^*(n) < p_M^*(n).$$

□

Proof of Proposition 2

Proof. First, we derive the conditions under which the equilibria in both modes are interior.

Under the pure marketplace mode, the equilibrium demand of each 3P seller is

$$D_M^*(n) = \frac{[1+(n-2)\gamma]\,(1-\tau-c)}{(1-\tau)\,[2+(n-3)\gamma]\,[1+(n-1)\gamma]}.$$

Therefore, this demand is strictly positive if and only if $c < 1-\tau$.

Under the hybrid mode, the equilibrium demands of the platform and each 3P seller are, respectively:

$$\hat{D}_P^*(n) = \frac{(1-c)AB - \tau(n-1)\gamma Q}{Q\Delta},$$
$$\hat{D}_S^*(n) = \frac{A\,[(1-\tau)B - cH]}{(1-\tau)Q\Delta},$$

where $A \equiv 1+(n-2)\gamma, B \equiv 2+(2n-3)\gamma, H \equiv 2+(2n-3+\tau)\gamma = B+\tau\gamma, Q \equiv 1+(n-1)\gamma$, and $\Delta \equiv 4+6(n-2)\gamma+\gamma^2\left[2n^2-9n+9-(n-1)\tau\right]$. All of these auxiliary quantities are strictly positive under the maintained parameter restrictions $n \geq 2$, $0<\gamma<1$, and $0<\tau<1$.

It follows that

$$\hat{D}_P^*(n) > 0 \quad \Longleftrightarrow \quad c < \hat{c}_P \equiv 1 - \frac{\tau(n-1)\gamma\,[1+(n-1)\gamma]}{[1+(n-2)\gamma]\,[2+(2n-3)\gamma]},$$
$$\hat{D}_S^*(n) > 0 \quad \Longleftrightarrow \quad c < \hat{c}_S \equiv \frac{(1-\tau)\,[2+(2n-3)\gamma]}{2+(2n-3+\tau)\gamma} = \frac{(1-\tau)B}{H}.$$

Because $\hat{c}_P - \hat{c}_S = \frac{\tau Q\Delta}{ABH} > 0$, we have $\hat{c}_S < \hat{c}_P$. Hence, the equilibrium under the hybrid mode is interior if and only if $c < \hat{c}_S$.

Comparing $p_S^*(n)$ and $p_M^*(n-1)$ yields

$$\hat{p}_S^*(n) - p_M^*(n-1) = \frac{\gamma(1-\gamma)K}{(1-\tau)\,[2+(n-4)\gamma]\,\Delta}\,(c-c_E)\,,$$

where $K \equiv 2(1-\tau)+\gamma\,[2n(1-\tau)+5\tau-3] > 0$ and $c_E \equiv \frac{(1-\tau)[2+\gamma(2n-3-(n-1)\tau)]}{K}$. Therefore, $\text{sign}\,[\hat{p}_S^*(n) - p_M^*(n-1)] = \text{sign}(c-c_E)$.

A direct comparison of $\hat{c}_S$, c_E and $1-\tau$ yields

$$\hat{c}_S < 1-\tau,$$
$$c_E - (1-\tau) = \frac{(1-\tau)\tau\,[2+(n-4)\gamma]}{K} > 0,$$

which implies that $\hat{c}_S < 1-\tau < c_E$. Thus, the condition $c < \hat{c}_S$ guarantees that the equilibria in both modes are interior and that $c < c_E$.

Equivalently, the interiority condition can be written as

$$0 < \tau < \hat{\tau} \equiv \frac{(1-c)\,[2+(2n-3)\gamma]}{2+(2n-3+c)\gamma}.$$

Therefore, over the relevant range of commission rates, $\tau < \hat{\tau}$, we have

$$\hat{p}_S^*(n) < p_M^*(n-1).$$

Hence, the platform's entry lowers the equilibrium prices charged by 3P sellers. □

Proof of Proposition 3

Proof. Substituting the linear demand system into the equilibrium conditions—Equation (1) for 3P sellers, $i = 2, \ldots, n$, and Equation (2) for the platform—yields the following result.

$$2\hat{p}_P^* - \delta(1+\tau_R)\sum_{i=2}^{n}\hat{p}_i^* = \frac{1-\gamma}{1+(n-2)\gamma} + c_P,$$

and

$$2\hat{p}_i^* - \delta\sum_{j\neq i}^{n}\hat{p}_j^* = \frac{1-\gamma}{1+(n-2)\gamma} + \frac{c_i}{1-\tau_D}, \qquad i \in \{2,\ldots,n\}.$$

Stacking the equilibrium pricing conditions yields

$$M(\tau_R)\hat{\mathbf{p}}^* = \frac{1-\gamma}{1+(n-2)\gamma}\mathbf{1}_n + C(\tau_D)\mathbf{c},$$

where

$$M(\tau_R) \equiv \begin{bmatrix} 2 & -\delta(1+\tau_R)\mathbf{1}_{n-1}^\top \\ -\delta\mathbf{1}_{n-1} & (2+\delta)I_{n-1} - \delta J_{n-1} \end{bmatrix}, \textit{and } C(\tau_D) \equiv \begin{bmatrix} 1 & \mathbf{0}_{n-1}^\top \\ \mathbf{0}_{n-1} & \frac{1}{1-\tau_D}I_{n-1} \end{bmatrix}.$$

Differentiating this equation with respect to the marginal-cost vector $\mathbf{c}$ yields

$$M(\tau_R)\rho(\tau_R, \tau_D) = C(\tau_D),$$

and hence

$$\rho \equiv \frac{\partial \hat{\mathbf{p}}^*}{\partial \mathbf{c}^\top} = M(\tau_R)^{-1}C(\tau_D).$$

$\square$

Proof of Proposition 4

Proof. Let $\kappa_D \equiv \frac{1}{1-\tau_D}$, $\ell \equiv 2-(n-2)\delta$, and $\Omega_R \equiv 2\ell-(n-1)\delta^2(1+\tau_R)$. The pass-through matrix from the platform's and 3P sellers' individual cost shocks to their equilibrium prices is given by

$$\rho = M(\tau_R)^{-1}C(\tau_D) = \begin{bmatrix} \dfrac{\ell}{\Omega_R} & \dfrac{\delta(1+\tau_R)\kappa_D}{\Omega_R}\mathbf{1}_{n-1}^\top \\ \dfrac{\delta}{\Omega_R}\mathbf{1}_{n-1} & \dfrac{\kappa_D}{2+\delta}I_{n-1} + \dfrac{\kappa_D}{n-1}\left[\dfrac{2}{\Omega_R} - \dfrac{1}{2+\delta}\right]J_{n-1} \end{bmatrix},$$

where $\tau_R = \tau_D = \tau$.

Therefore, the four blocks of the pass-through matrix contain five distinct types of pass-through rates. The upper-left block of the pass-through matrix $\rho_{PP} \equiv \frac{\partial \hat{p}_P^*}{\partial c_P} = \frac{\ell}{\Omega_R}$ denotes the platform's own-cost pass-through rate. For any $S \in \{2,\ldots,n\}$, the upper-right block contains $\rho_{PS} \equiv \frac{\partial \hat{p}_P^*}{\partial c_S} = \frac{\delta(1+\tau_R)\kappa_D}{\Omega_R}$, which denotes the pass-through of a 3P seller's cost shock to the platform's price. The lower-left block, by contrast, contains $\rho_{SP} \equiv \frac{\partial \hat{p}_S^*}{\partial c_P} = \frac{\delta}{\Omega_R}$ and denotes the pass-through of the platform's cost shock to a 3P seller's price. The lower-right block concerns only the 3P sellers. A 3P seller's own-cost pass-through rate is $\rho_{SS} \equiv \frac{\partial \hat{p}_S^*}{\partial c_S} = \frac{\kappa_D}{2+\delta} + \frac{\kappa_D}{n-1}\left[\frac{2}{\Omega_R} - \frac{1}{2+\delta}\right]$. For any distinct $S, S' \in \{2,\ldots,n\}$, $\rho_{SS'} \equiv \frac{\partial \hat{p}_S^*}{\partial c_{S'}} = \frac{\kappa_D}{n-1}\left[\frac{2}{\Omega_R} - \frac{1}{2+\delta}\right]$ denotes the pass-through of one 3P seller's cost shock to another 3P seller's price.

Thus, the pass-through rates of the common cost shock to the platform's price and to a 3P seller's price are, respectively,

$$\rho_P = \rho_{PP} + \sum_{S=2}^{n} \rho_{PS} = \frac{\ell + (n-1)\delta(1+\tau_R)\kappa_D}{\Omega_R},$$

and

$$\rho_S = \rho_{SS} + \rho_{SP} + \sum_{\substack{S'=2 \\ S' \neq S}}^{n} \rho_{SS'} = \frac{2\kappa_D + \delta}{\Omega_R}, \qquad S \in \{2, \ldots, n\}.$$

We now impose $\tau_R = \tau_D = \tau \in (0,1)$, so that $\kappa_D = 1/(1-\tau)$. Under the maintained assumptions on the demand system, $0 < \delta < 1/(n-1)$, and hence $\Omega_R > 0$.

For own-cost pass-through, direct comparison gives

$$\rho_{SS} - \rho_{PP} = \frac{2\tau\left[2 - (n-3)\delta - (n-2)\delta^2\right]}{(1-\tau)(2+\delta)\Omega_R} > 0.$$

The inequality follows from $\tau > 0$, $\Omega_R > 0$, and $0 < \delta < 1/(n-1)$, which also implies $2 - (n - 3)\delta - (n-2)\delta^2 > 0$. Therefore, $\rho_{PP} < \rho_{SS}$, establishing part (1).

For cross-cost pass-through, when $n \geq 3$, there exist two distinct 3P sellers S and S'. We have

$$\rho_{SS'} - \rho_{SP} = \frac{2\delta\tau(1+\delta)}{(1-\tau)(2+\delta)\Omega_R} > 0$$

and

$$\rho_{PS} - \rho_{SS'} = \frac{2\delta\tau}{(1-\tau)(2+\delta)\Omega_R} > 0.$$

Hence, $\rho_{SP} < \rho_{SS'} < \rho_{PS}$, which establishes part (2).

Finally, comparing the pass-through rates of an industry-wide cost shock gives

$$\rho_S - \rho_P = \frac{2\tau\left[1 - (n-1)\delta\right]}{(1-\tau)\Omega_R} > 0,$$

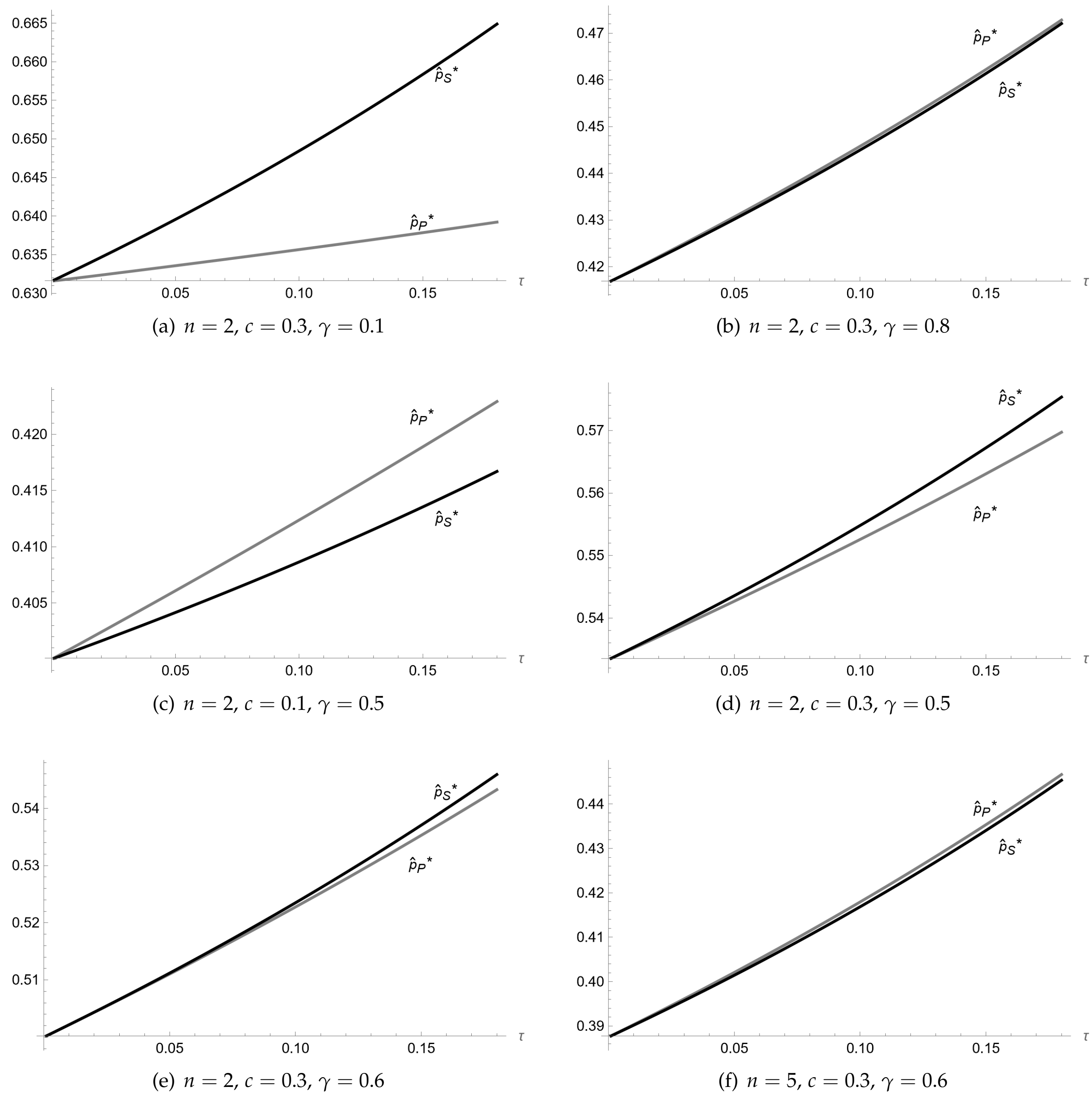
where the inequality follows from $0 < \delta < 1/(n-1)$. Thus, $\rho_P < \rho_S$, establishing part (3). □

A.2 Numerical Illustrations of Proposition 1

As stated in Proposition 1, the comparison between $\hat{p}_P^*(n)$ and $\hat{p}_S^*(n)$, i.e., the two equilibrium prices on the hybrid platform, is equivalent to the comparison between the platform's entry effect, $\hat{p}_S^*(n) - p_M^*(n-1)$, and a 3P seller's entry effect, $p_M^*(n) - p_M^*(n-1)$. The numerical examples shown in Figure A.1 therefore focus on the comparison between $\hat{p}_P^*(n)$ and $\hat{p}_S^*(n)$ to illustrate how the equilibrium price ranking varies across different parameter configurations.

In panels (a) and (b) of Figure A.1, n and c are held fixed, while γ increases from 0.1 to 0.8.

Figure A.1: Comparison of semi-symmetric equilibrium prices under hybrid mode across parameter configurations



(a) $n = 2$, $c = 0.3$, $\gamma = 0.1$

(b) $n = 2$, $c = 0.3$, $\gamma = 0.8$

(c) $n = 2$, $c = 0.1$, $\gamma = 0.5$

(d) $n = 2$, $c = 0.3$, $\gamma = 0.5$

(e) $n = 2$, $c = 0.3$, $\gamma = 0.6$

(f) $n = 5$, $c = 0.3$, $\gamma = 0.6$

Notes: The figure plots the semi-symmetric equilibrium prices under hybrid mode as functions of the commission rate τ across different parameter configurations. The gray line represents the platform's equilibrium price, $\hat{p}_P^*(n)$, and the black line represents the equilibrium price of 3P sellers, $\hat{p}_S^*(n)$. The values of n, c, and γ are reported in the title of each panel. When $\hat{p}_P^*(n) > \hat{p}_S^*(n)$, the revenue-sharing effect dominates the effect associated with eliminating the double-markup distortion; when $\hat{p}_P^*(n) < \hat{p}_S^*(n)$, the latter dominates.

When γ is relatively low, as in panel (a), the effect associated with eliminating the double-markup distortion dominates, so that the platform charges a lower price than 3P sellers. When γ is sufficiently high, as in panel (b), the revenue-sharing effect dominates, so that the platform charges a higher price than 3P sellers. Intuitively, a larger γ makes the platform's product and the 3P products closer substitutes. As a result, an increase in the platform's own price shifts more demand toward 3P sellers. Because the platform earns commission revenue from those sales, it internalizes this demand diversion and therefore has a stronger incentive to raise its own price. This can be seen directly from the platform's first-order condition:

$$D_P + (p_P - c)\frac{\partial D_P}{\partial p_P} + \tau \sum_{i=2}^{n} p_i \frac{\partial D_i}{\partial p_P} = 0.$$

The last term captures the revenue-sharing effect. Under the linear demand system, the aggregate diversion ratio is

$$D \equiv \left| \frac{\sum_{i=2}^{n} \frac{\partial D_i}{\partial p_P}}{\frac{\partial D_P}{\partial p_P}} \right| = \frac{(n-1)\gamma}{1+(n-2)\gamma},$$

which is increasing in γ. Thus, as products become more substitutable, a larger share of the demand lost by the platform when it raises its price is captured by 3P sellers. This makes the associated increase in commission revenue more important relative to the platform's loss of retail profit.

Then, in panels (c) and (d), n and γ are held fixed, while c increases from 0.1 to 0.3. When c is relatively low, as in panel (c), the revenue-sharing effect dominates, so that the platform charges a higher price than 3P sellers. When c is sufficiently high, as in panel (d), the effect associated with eliminating the double-markup distortion dominates, so that the platform charges a lower price than 3P sellers. Intuitively, as c increases, the double-markup distortion faced by 3P sellers becomes more pronounced, making its elimination relatively more important for the platform's pricing decision.

Finally, in panels (e) and (f), c and γ are held fixed, while n increases from 2 to 5. When n is relatively small, as in panel (e), the effect associated with eliminating the double-markup distortion dominates, so that the platform charges a lower price than 3P sellers. When n is sufficiently large, as in panel (f), the revenue-sharing effect dominates, so that the platform charges a higher price than 3P sellers. Intuitively, as the number of 3P sellers increases, the platform internalizes the effect of its own price on a larger set of commission-generating sales. When the platform raises its price, demand is diverted toward the $n-1$ 3P sellers, increasing the commission revenue it earns from their sales. Consistent with this intuition, the platform's aggregate diversion ratio D is increasing in n. Thus, with more competitors, a larger share of the demand lost by the platform following a price increase is captured by commission-paying 3P sellers, making the revenue-sharing effect relatively more important.

B Data Construction and Measurement

B.1 Keepa Data: Collection and Processing

As a first step, we use Keepa's "Product Finder" tool to arrive at a list of ASINs related to a given product category (e.g., Coffee or Men's T-Shirts). Table B.1 shows the query parameters used for each product category queried. Note that a product (ASIN) on Amazon can belong to one *Root Category* and multiple *Browse Nodes*[21]. The latter are called *Subcategories* on the Keepa website. We configure the Keepa Product Finder filters so that only subcategories directly related to our target product (e.g., "Ground Coffee") are included, while certain adjacent but irrelevant subcategories (e.g., "Instant Coffee") are explicitly excluded. This enables us to focus on truly similar products.

Table B.1: Query parameters used in initial product data collection process on Keepa

Root Category and Product Category	Included and excluded subcategories
Root Category: Grocery & Gourmet Food	
Coffee	· include: Ground Coffee · exclude: Tea; Instant Coffee
Cocoa	· include: Cocoa · exclude: Chocolate Drink Mixes; Hot Chocolate & Malted Drinks; Thanksgiving, Tea & Cocoa
Pasta	· include: Pasta & Noodles · exclude: Fresh Pasta & Sauces; Gnocchi
Root Category: Electronics	
TVs	· include: LED & LCD TVs · exclude: Remote Controls; TV Mounts, Stands & Turntables; Satellite Equipment
USB Hubs	· include: USB Hubs
Micro SD Cards	· include: Micro SD Cards
Root Category: Appliances	
Washing Machines	· include: Washers · exclude: Countertop Dishwashers; Built-in Dishwashers; Portable Dishwashers; Washer Parts & Accessories; Portable Washers
Root Category: Clothing, Shoes & Jewelry	
Men's T-Shirts	· include: Men's T-Shirts · exclude: Tights, Pants & Shorts; Leggings

For product categories with over 10,000 products listed on `amazon.us`, we only collect data on those 10,000 products with the lowest current sales rank or the lowest 180-day average sales rank at the time of scraping, thereby focusing our attention on products that actually generate sales and

[21] See `https://webservices.amazon.com/paapi5/documentation/use-cases/organization-of-items-on-amazon/browse-nodes/browse-nodes-and-items.html` (accessed on 25/11/2024). The Browse Node IDs of the subcategories listed in Table B.1 are not displayed for readability reasons, but can be provided upon request.

are (normally) in stock. After obtaining a list of ASINs for a given product category, we query these ASINs one by one using "Product Object" queries[22] through Keepa's Application Programming Interface (API), and thereby obtain product and price information. We query all categories at multiple points in time in the period of July 2024 through March 2026.

After collecting the list of ASINs, we exclude products which (either based on manual inspection or based on the variable "type") seem to belong to a different category and are thus misclassified. For products in the Pasta category, we exclude products marked with terms like "buckwheat", "rice" or "gluten-free", so as to focus on products made from wheat; for products in the coffee category, we exclude chicory or instant coffee sticks, for instance. Within a given category, we only use data on products that are ever on sale on `amazon.com` and thus have a price indicated at any point since 2021.

The original "Product Object" data downloaded through the Keepa API essentially consists of separate datasets for the lowest "marketplace" offer (which can be Amazon or a 3P seller's price, called NEW in Keepa); the lowest price set by any 3P seller operating under FBA; the lowest price set by any 3P seller operating under FBM; Amazon's price; and the price of the seller whose offer is displayed in the Buy Box. 3P FBM prices are inclusive of shipping costs that consumers will need to pay, while all other price measures do not include shipping costs (and are eligible for free shipping for *Amazon Prime* subscribers). (It is worth noting that, prior to 2026, Keepa updated the lowest prices under FBA/FBM at slightly lower frequencies than the other prices.) Each original-data row corresponds to a given price change at a given moment in time (a price becoming unavailable or available corresponds to a "change" as well). We define the lowest 3P price (which we call NEW_third) as the marketplace price (NEW) whenever it differs from the Amazon price, and as the minimum of the lowest FBA and FBM prices otherwise.

Adjustment for occasionally occurring delayed Amazon price recording. The product-level price data from Keepa occasionally exhibits a timing discrepancy in which Amazon's offer is detected with a short lag. Specifically, when a previously out-of-stock Amazon offer returns to the marketplace, Keepa's system may register the resulting change in the marketplace price (NEW) and seller count (COUNT_NEW) before identifying the Amazon offer itself. During this interim period – lasting only a few days – the AMAZON price field remains missing even though the marketplace price already reflects Amazon's presence, which would cause the lowest 3P price (NEW_third) to

[22] See `https://keepa.com/#!discuss/t/product-object/116` (last accessed 18/02/2026).

be incorrectly attributed. We confirmed this data-collection artifact directly with the Keepa team[23]. To address it, we apply a conservative correction procedure that detects instances in which (i) the seller count increases by exactly one, (ii) the marketplace price simultaneously decreases, (iii) the Amazon price is not yet recorded, (iv) no FBA or FBM price equals the marketplace price on the affected days (ruling out the possibility that the price drop is attributable to a 3P seller), and (v) within seven days the Amazon price appears while the seller count remains *unchanged*. When all five conditions are met, we backfill the Amazon price to the date of the initial seller-count increase. This adjustment affects a small number of product-days (0.001% in the case of Cocoa and 0.002% in the case of coffee, for instance) and ensures that Amazon's entry date is correctly recorded.

Addressing a small number of apparent pricing errors. We observed a small number of apparent pricing errors in the original data, i.e., instances where a product's price jumps up by several multiples of the original price for a period of time. For instance, a package of coffee bearing the ASIN B09B8TP9RQ, typically priced at around $24.99, is suddenly offered at $5,889.34 from February 20 to February 26, 2025 – an over 200-fold increase in its price – before coming back down. These may result from data-entry errors or algorithmic malfunctions. We never observe such patterns in the case of Amazon's offers, only for 3P sellers' prices. To detect these upward price deviations in a conservative but sensible way and exclude them from our dataset upfront, we employ a multi-stage procedure to identify these errors at both the observation and product levels. For each ASIN, we calculate the interquartile range (IQR = Q75 - Q25) of observed prices in a given category. An individual price observation is flagged as an outlier if it satisfies all three conditions: (1) Price $>$ Q75 + 30 × IQR; (2) Price - Q75 $>$ 2 × median_all, where median_all is the median price across all products in that category; and (3) Price > 2 × median_product. These criteria jointly ensure that flagged observations represent extreme deviations in both product-specific terms (IQR criterion), absolute terms (preventing spurious flags in low-variance products), and relative terms (substantial deviation from the product's typical price level). While we allow for the exclusion of entire *products* from the sample when outlier prevalence suggests systematic data quality issues[24], in the case of coffee or cocoa, there are no such systematic data quality issues for any products in our sample. More generally, only around 0.2% of all ASINs are affected by these pricing errors. Finally, in the coffee category, we exclude one brand ("San Marco Coffee") which operated under

[23]See https://keepa.com/#!discuss/t/price-update-frequency-for-amazon/18645 (last accessed 18/02/2026).

[24]Specifically, we remove all observations for a product if either: (1) more than 30% of observed prices are flagged as outliers, or (2) the product has at least 10 outliers and fewer than 50 total price observations.

a dropshipping arrangement until 2024. For this brand, the data show an extreme repricing event, followed by a synchronized cessation of Amazon's offers for all products from that brand. We contacted the brand and obtained confirmation that a termination of the vendor relationship had taken place, which is consistent with the observed pricing patterns.

After creating a daily panel of products' prices (recorded at midnight of a given date), we further exclude products if their average 3P (Amazon) price, standardized by weight when considering grocery products, across all observed days lies above the 98th, or below the 2nd percentile of all 3P sellers' (Amazon's) daily prices. Very often these products' category in fact seems to have been misclassified, so that these products should not be part of the sample.

B.2 Construction of Product Weight Measures

For our analysis, a precise extraction of product package sizes is crucial. However, package sizes from the relevant variables included by Keepa are missing for many of the products. Instead, package sizes are often accurately described in product titles, but due to nonstandardized formatting, are difficult to extract using standard string-matching methods. To address this, we employ OpenAI's gpt-4o-mini model. The prompt instructs the model to identify and compute information on total weight or volume of the content as indicated in the product title, and is displayed below. Note that an ASIN's weight cannot vary over time; the introduction of a different package size should generally imply a new ASIN.

```
You are tasked with extracting the total weight or volume of a grocery
   product from its description. Follow these specific rules:
    1. **Accurate extraction only**: Use only the information explicitly
       provided in the product description. Do not infer or assume any
       details.
    2. **Output format**: Provide the result as ‘number|unit‘.
       - If the description includes a total weight or volume (including
          multipacks), calculate the combined total if necessary.
       - Units like ounces (oz), grams (g), pounds (lb), kilograms (kg),
          liters (L), or milliliters (ml) should be preserved as stated.
    3. **No weight or volume available**: If no weight or volume information
        is provided, return ‘NA|NA‘.
```

```
**Examples**:
- **Input**: "Starbucks Caffe Verona Coffee, Dark, Ground, 12-Ounce Bags
    (Pack of 3)"
  **Output**: 36|oz
- **Input**: "Four Sigmatic Think Mushroom Coffee | Organic Ground
   Coffee with Lion's Mane Mushroom and Chaga Mushroom | 12oz Bag"
  **Output**: 12|oz
- **Input**: "AmazonFresh Go For The Bold Ground Coffee, Dark Roast"
  **Output**: NA|NA
- **Input**: "AmazonFresh French Vanilla Flavored Coffee, Ground, Medium
    Roast, 500g"
  **Output**: 500|g
Now, extract the total weight or volume from the following description:
---
{text}
---
```

To reduce output variation, we set the temperature parameter to 0. We moreover include a system message – "role": "system", "content": "You are an expert at analyzing grocery product descriptions and extracting accurate weight or volume information. You strictly follow instructions, avoid assumptions, and format answers as specified." – which instructs the model to follow the extraction protocol and avoid unsupported assumptions.

Table B.2 displays the resulting distribution of weights across ASINs, showing that we are able to recover the product weight for a vast majority of products, and showing that the median package size is relatively standard.

Table B.2: Distribution of product weights across ASINs in grocery categories

Category	ASINs	ASINs w/ weight available	Mean	Median	SD	P10	P90
Cocoa	1,432	1,393	1137.2	453.6	2628.3	170.5	2268.0
Coffee	14,532	14,282	828.1	453.6	894.2	270.0	2041.2
Pasta	7,709	7,477	1861.2	997.9	2258.4	340.2	5000.0

Note: The table reports ASIN-level weight statistics (in grams) for Cocoa, Coffee, and Pasta products.

B.3 Building a Panel of Amazon Marketplace Data

Sales rank. Given that a product's sales rank is not always recorded on a daily basis, we forward-fill this variable, so that a product's sales rank on a given day reflects the latest sales rank that was recorded on Keepa.

Defining Amazon "entry" and "exit". We define Amazon entry to be the first day on which Amazon's price variable becomes available, following a day at which it was not available. Likewise, we define exit as the first day on which Amazon's price variable becomes unavailable, following a day of Amazon's price being available.

Identifying Amazon private-label products. While not core to our analysis, we report the percentage of Amazon private-label products in Table 1. We identify these private label products by tagging products whose manufacturer is listed as "365 by Whole Foods Market," "Amazon.com Services, Inc.," "Amazon.com Services LLC.," or "Amazon," as well as products whose brand is listed as "Amazon Fresh" or "Amazon Basics."

Building a weekly panel. From a daily panel of products with prices recorded at midnight, we build a weekly panel by taking the average across all observed prices and sales rank. If a price variable is not available on some days of a given week, we take the average across the remaining days on which it is observed. While our analysis covers January 2021 to March 2025, note that, in order to generate the pass-through results, we take lagged variables going back into 2020.

Identifying higher-turnover products. We use the sales rank variable to identify relatively high-turnover products: we call a product "popular" if it is consistently highly-ranked (which is defined as its sales rank being below the median of the overall sales rank distribution in at least 50% of its observations). We also consider a product's percentile in the sales-rank distribution.

B.4 Commodity Price Data

Cocoa beans. The cocoa bean price-series we use stems from the International Cocoa Organization (ICCO) and is available via www.icco.org/statistics. We use the ICCO daily price for cocoa beans in US$, defined as "the average of the quotations of the nearest three active futures trading months on ICE Futures Europe (London) and ICE Futures US (New York) at the time of London close". We refer interested readers to the website for further information.

Arabica coffee beans. Daily pricing data on Arabica coffee beans are obtained from Trading Economics, available on tradingeconomics.com/commodity/coffee. The data corresponds to Arabica Coffee C Futures contracts in US$ that trade on the Intercontinental Exchange (see https://www.ice.com/products/15/Coffee-C-Futures), and is the global benchmark for Arabica Coffee. Because Arabica and Robusta coffee-bean prices tend to co-move, and because Arabica accounts for roughly 90% of U.S. coffee imports,[25] we use the Arabica coffee price as a proxy for the commodity input cost of ground coffee sold in the United States.

For both of these commodity price series, we create weekly average prices by averaging across all observed prices in a given week. Figure 7 shows the weekly time series of both prices.

Commodity-Price Conversion Factors. For the levels regressions, we need to express each raw-commodity price in downstream-product-equivalent units (Sangani, 2026). Let c_{jt}^{raw} denote the raw-commodity price per gram in category j, and let m_j denote the adopted raw-commodity-equivalent factor for one gram of the downstream product. We define $c_{jt} = m_j c_{jt}^{\text{raw}}$ and use Δc_{jt} in the distributed-lag regressions. Thus, $\sum_{k=0}^{K} \beta_k$ measures the cumulative retail-price response to a change in the converted commodity input-price measure, with both variables expressed per gram of downstream product. The factors and their sources are reported in Table B.3. (Note that the conversion rescales level coefficients and standard errors proportionally and therefore leaves t-statistics and the within-category Amazon–3P comparison unchanged.) The coffee factor is a processing-yield adjustment (weight is lost during roasting). For cocoa, the factor is conceptually different: it measures the beans processed per unit of powder, but the same beans also produce cocoa butter.

Table B.3: Commodity-price conversion factors

Downstream product category	Conversion factor m_j
Ground roasted coffee	1.235
Baking cocoa (cocoa powder)	2.44

Notes: Raw-commodity prices are multiplied by m_j. For coffee, $m_{\text{coffee}} = 1/(1-0.19) = 1.235$, using the 19 percent roasting-weight loss in Nakamura and Zerom (2010) and Sangani (2026). For cocoa, $m_{\text{cocoa}} = 1/(0.82 \times 0.50) = 2.44$, using the mass-balance conversion ratios reported by Fairtrade International (2026, Requirement 2.1.5). The same ratios are also used by Rainforest Alliance (2025, Section 5).

[25] See https://www.ers.usda.gov/data-products/charts-of-note/110079? (accessed 13/07/2026).

C Additional Stylized Facts

C.1 Additional Facts on Amazon's Hybrid Platform

Survey of previous literature on Amazon reselling activity. Several academic papers shed light on this question by studying the predictors of Amazon's decision to directly resell a product (i.e., to "enter"). Hagiu and Wright (2015) use data on DVDs that were sold on Amazon in 2014, and find that products offered by Amazon have lower sales rank, i.e., are more popular, than those offered by 3P sellers. Zhu and Liu (2018) employ data from 2013 and 2014 and finds that Amazon is more likely to begin reselling products that experience higher sales and have better reviews, and which are not yet using Amazon's fulfillment services. A working paper by Crawford et al. (2022) uses proprietary data on prices and revenues in Amazon's Home & Kitchen category. The paper focuses on Amazon's decision to begin selling products for which only a single 3P merchant supplied that product in the first four weeks of the product's existence. The authors find that Amazon starts reselling products that experience high demand growth and where competition is low (i.e., few active sellers are present). This is true both relative to products which Amazon does *not* enter, and relative to the entry decisions by high-revenue 3P sellers. Entry by Amazon occurs even if a large or efficient 3P seller is present. Amazon is also more likely to enter when 3P listings exist but generate no sales, suggesting the presence of low-quality or capacity-constrained sellers. The authors argue that these patterns suggest the internalization of platform externalities as a possible motive for Amazon's entry decisions. Among those externalities are expanding product variety, ensuring product availability, and mitigating seller market power.

Concerning Amazon's presence as a reseller more generally, the authors find that nearly half of the revenue from products in which Amazon is a reseller comes from cases of *de novo* entry – products that had not been sold by any 3P seller before Amazon's entry as a reseller. Amazon's private label products play a minor role in the Home & Kitchen category, accounting for only 0.3% of total revenues. Amazon is present in products accounting for 39% of revenues in the Home & Kitchen category.

The findings by Bennati (2026), who studies the market for headphones and uses sales estimates stemming from the market intelligence company AmzScout, are very much in line with this. The author's findings suggest that 3P sellers concentrate on products in the long tail, whereas Amazon resells products that are in higher demand and of higher quality.

Correlates of Amazon reselling activity. Taking the example of coffee, we find that, in the cross-section, the brand and the manufacturer are the main factors correlated with Amazon's entry patterns. Controlling for brand and manufacturer, we moreover find that products where Amazon is a reseller at any point during our period of observation are likely to derive more sales, and are likely to feature more other sellers, as can be seen in Table C.1. The finding that more competition by 3P sellers is correlated with Amazon reselling the product could in fact be driven by unobserved (by the econometrician) demand shocks, which might cause entry by both Amazon and other merchants. Hence, our measure of "high competition" might in fact be a proxy for "high demand". (Of course, this result can by no means be interpreted causally, and is purely descriptive.)

In the time series, interestingly we find that, across our product categories, the overall number of products resold by Amazon has been constant over the years, while the number of products resold by 3P sellers has increased over time (not shown in paper).

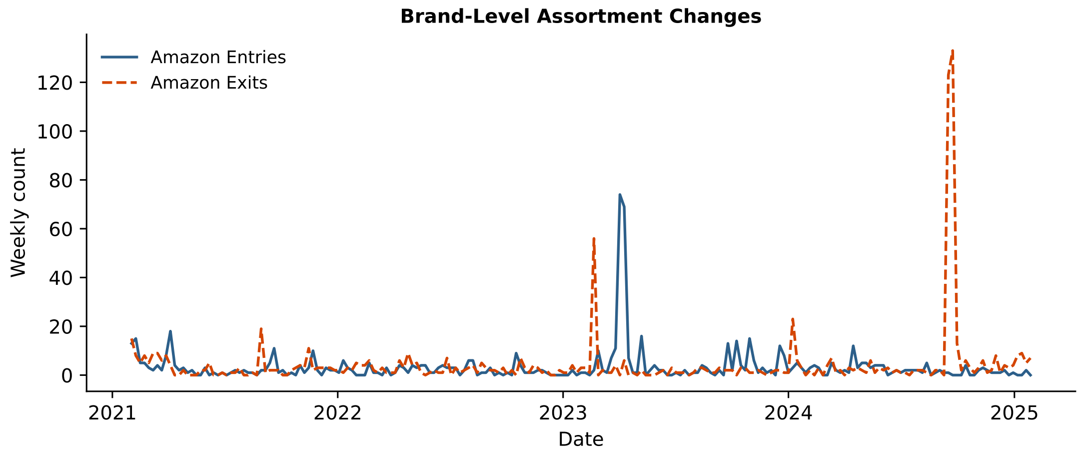
Table C.1: Correlates of whether a product is ever supplied by Amazon, Coffee products only

	Dependent variable:	
	$\mathbb{1}${product ever sold by Amazon}	
1{popular}	0.123**	0.065**
	(0.052)	(0.028)
1{has any reviews}	0.209***	0.036
	(0.070)	(0.031)
avg number of 3P sellers	0.008	0.008***
	(0.014)	(0.003)
package size (weight, in kg)	0.000	0.000
	(0.00000)	(0.000)
product was ever stocked out	0.163***	0.013
	(0.058)	(0.031)
Manufacturer FE		✓
Brand FE		✓
Observations	12,432	12,342
Adjusted R^2	0.063	0.784
SEs clustered at manufacturer level.	*p<0.1; **p<0.05; ***p<0.01	

Notes: A cross-sectional dataset was created by beginning with a daily panel ranging from January 1st 2021 to March 31 2025, and collapsing it to the ASIN-level so that each observation corresponds to an ASIN. 1{popular} indicates whether a product is "popular" using the popularity measure defined in Section B. The average number of 3P sellers is defined as an average across all days on which a product is available for sale. The coefficients from a linear regression are shown.

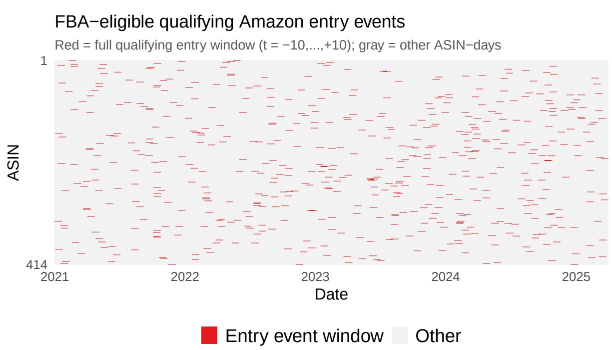
C.2 Additional Facts on Amazon's Intermittent Retail Availability

Figure C.1: Amazon assortment change on the brand level: case of "The Coffee Fool"

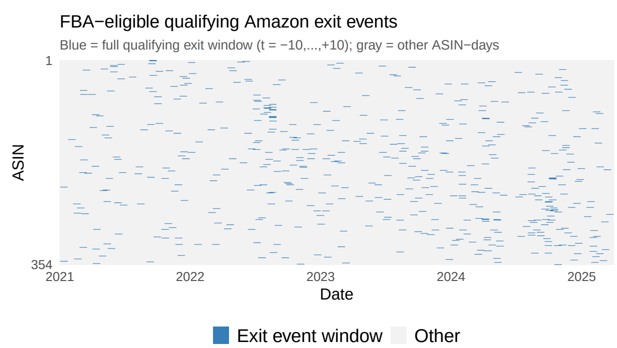


Notes: To illustrate that Amazon sometimes enters or exits systematically many ASINs of a particular brand, the figure shows the weekly number of (re-)entries/exits of Amazon over time in the coffee category, focusing on the brand "The Coffee Fool" only. We only count a re-entry (re-exit) as such if it is followed by a spell of presence (absence) of 5 days or more.

Figure C.2: Amazon's presence at product level: Entry and exit events



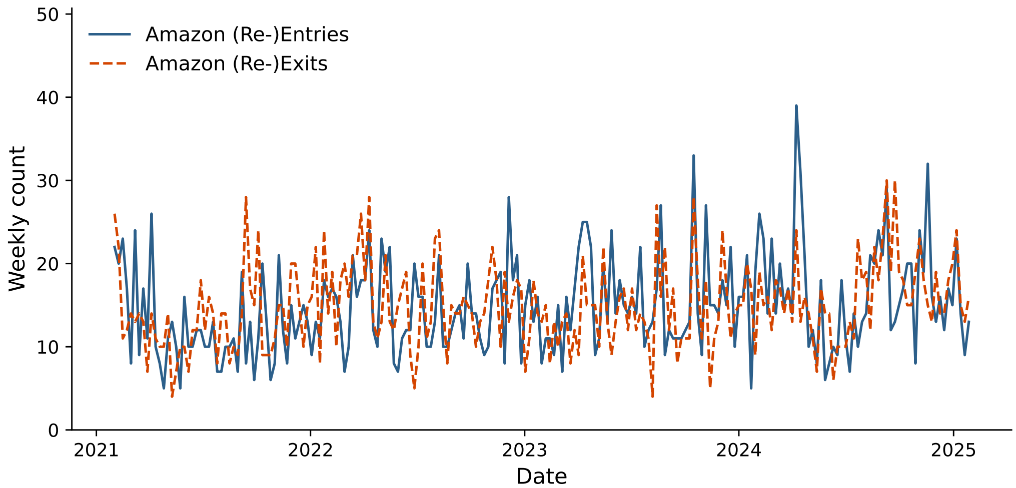
(a) Entry events.



(b) Exit events.

Notes: The figures plot qualifying Amazon entry/exit event windows over the sample period for the set of ASINs sold by both Amazon and 3P sellers. Each row corresponds to an ASIN, and ASINs are sorted vertically by the share of days on which Amazon is present in (the ASIN which Amazon is present least often is displayed at the top). Panel (a) shows entry events, with red bars indicating qualifying event windows around Amazon entry. Panel (b) shows exit events, with blue bars indicating qualifying event windows around Amazon exit.

Figure C.3: Recurring Amazon entries and exits over time: Pasta category



Notes: The figure shows the analogous to Figure 2, but this time focusing on our data on Pasta. The lines indicate the weekly number of re-exits and re-entries by Amazon over time for the Pasta category. We exclude brands that account for more than 20% of aggregate Amazon entry or exit flows in the ten highest-activity weeks (two brands in the case of pasta), as these spikes reflect systematic brand-level assortment changes rather than product-level entry or exit decisions. We only count a re-entry (re-exit) as such if they are followed by a spell of presence (absence) of 5 days or more (this is robust to other cutoffs).

Table C.2: Descriptives on Amazon retail presence spell duration among switcher ASINs (FBA-active days)

	N	Mean	Std. Dev.	Median	P10	P90
Panel A: Spell-level distribution (spell duration in days)						
Amazon present	8,782	31.33	88.39	3.00	1.00	82.00
Amazon absent	9,010	65.90	157.59	9.00	1.00	173.00
Panel B: Per-ASIN number of spells						
Amazon present	1,159	7.58	11.56	3.00	1.00	18.00
Amazon absent	1,159	7.77	11.63	4.00	1.00	18.00
Panel C: Per-ASIN mean spell duration (days)						
Amazon present	1,159	65.17	93.54	34.10	4.16	156.60
Amazon absent	1,159	148.17	204.55	61.50	6.00	427.30

Notes: Coffee category; ASIN-day panel (midnight prices), January 2021–March 2025. ASINs are restricted to those satisfying the event-study screen: at least 10 consecutive days with both Amazon and the lowest 3P price (NEW_third) non-missing, and at least 10 consecutive days with only the lowest 3P price observed (and Amazon missing). We further restrict to switcher ASINs in the 3P-active sample (days with FBA observed): at least one day where Amazon is present and one day where it is absent. A spell is a maximal run in which Amazon's presence status is unchanged.

Table C.3: Predictors of Amazon entry using Logit models (3P-only days at risk, using FBA prices)

Dependent variable	Qualifying Amazon entry ($t+1$) event (continuous 3P availability under FBA)			
	(1) Baseline	(2) + 3P price	(3) + Sales rank	(4) + lagged changes
Num. 3P sellers ($t-1$)	0.0400***	0.0417***	0.0412***	0.0348**
	(0.0140)	(0.0143)	(0.0144)	(0.0161)
weekdayTue	0.4278**			
	(0.1845)			
weekdayWed	0.7976***			
	(0.1777)			
weekdayThu	0.6070***			
	(0.1821)			
weekdayFri	0.5886***			
	(0.1830)			
weekdaySat	0.4593**			
	(0.1864)			
weekdaySun	0.0822			
	(0.1976)			
FBA price ($t-1$)		0.0090	0.0107	0.0228***
		(0.0111)	(0.0107)	(0.0083)
log Sales rank ($t-1$)			-0.0730*	-0.0961**
			(0.0396)	(0.0444)
Δ_3 log Sales rank ($t-1$)				-0.0508
				(0.0701)
Δ Num. 3P sellers ($t-1$)				0.1874***
				(0.0537)
Δ FBA price ($t-1$)				-0.0062
				(0.0167)
Δ FBM price ($t-1$)				0.0163
				(0.0139)
ASIN	✓	✓	✓	✓
Week	✓	✓	✓	✓
Observations	227,755	227,755	224,960	151,766

Notes: The estimation sample consists of all ASIN-days at risk of entry, that is, every 3P-only day (Amazon absent) for switcher ASINs (ASINs that ever switch between a "dual" and a 3P-only regime). The dependent variable equals one if a baseline qualifying Amazon entry – requiring continuous 3P (under FBA) availability throughout the event window – occurs on the following day, and zero otherwise. Models are estimated by discrete-time logit. Δ_3 log Sales rank ($t-1$) is the three-day change in the natural log of the Amazon sales rank measured at $t-1$, i.e. $\log(\text{rank}_{t-1}) - \log(\text{rank}_{t-4})$; it approximates the three-day percentage change in sales rank. Because sales rank is inversely related to units sold, a positive value indicates a deterioration (decline) in the product's relative sales performance. Clustered (ASIN) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table C.4: Predictors of Amazon exit using Logit models (both-regime days at risk, using FBA prices)

Dependent variable	Qualifying Amazon exit ($t+1$) event (continuous 3P availability under FBA)			
	(1) Baseline	(2) + Prices	(3) + Sales & buy box	(4) + Amazon higher + lagged changes
Num. 3P sellers ($t-1$)	0.0455**	0.0454**	0.0464**	0.0417**
	(0.0194)	(0.0193)	(0.0188)	(0.0205)
weekdayTue	0.1438			
	(0.1663)			
weekdayWed	0.2188			
	(0.1729)			
weekdayThu	-0.0301			
	(0.1750)			
weekdayFri	0.1057			
	(0.1742)			
weekdaySat	0.2361			
	(0.1659)			
weekdaySun	0.0168			
	(0.1863)			
Amazon price ($t-1$)		-0.0271	-0.0227	-0.0152
		(0.0188)	(0.0196)	(0.0218)
FBA price ($t-1$)		0.0011	-0.0045	-0.0031
		(0.0104)	(0.0119)	(0.0120)
log Sales rank ($t-1$)			0.1307**	0.1331**
			(0.0578)	(0.0646)
Amazon wins buy box ($t-1$)			0.2608	0.4202*
			(0.1766)	(0.2358)
Δ_3 log Sales rank ($t-1$)				0.0384
				(0.1449)
Amazon > 3P price ($t-1$)				0.1232
				(0.2079)
Δ Num. 3P sellers ($t-1$)				0.1239
				(0.0857)
Δ FBA price ($t-1$)				-0.0021
				(0.0278)
Δ FBM price ($t-1$)				0.0024
				(0.0062)
Δ Amazon price ($t-1$)				-0.0080*
				(0.0046)
ASIN	✓	✓	✓	✓
Week	✓	✓	✓	✓
Observations	73,220	73,220	72,387	46,037

Notes: The estimation sample consists of all ASIN-days at risk of exit, that is, every day with Amazon present for switcher ASINs (ASINs that ever switch between a "dual" and a 3P-only regime). The dependent variable equals one if a baseline qualifying Amazon exit – requiring continuous 3P (under FBA) availability throughout the event window – occurs on the following day, and zero otherwise. Models are estimated by discrete-time logit. Δ_3 log Sales rank ($t-1$) is the three-day change in the natural log of the Amazon sales rank measured at $t-1$, i.e. $\log(\text{rank}_{t-1}) - \log(\text{rank}_{t-4})$; it approximates the three-day percentage change in sales rank. Because sales rank is inversely related to units sold, a positive value indicates a deterioration (decline) in the product's relative sales performance. Clustered (ASIN) standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table C.5: Descriptives: Seller regimes (FBA as 3P price)

	Obs.	Share	N (ASINs)
Panel A: ASIN-day regime counts			
3P only	641,316	0.264	1,280
Amazon only	695,212	0.286	1,420
Both	287,892	0.119	1,283
Neither	802,895	0.331	1,559
Total	2,427,315	1.000	1,565

	Mean	Median	P10	P90
Panel B: Within-ASIN share of days in regime				
3P only	0.264	0.145	0.000	0.727
Amazon only	0.286	0.255	0.001	0.681
Both	0.119	0.055	0.000	0.325
Neither	0.331	0.245	0.029	0.756

	N (spells)	Mean duration	Median duration
Panel C: Regime persistence (spell duration in days)			
3P only	15,017	42.71	11.00
Amazon only	12,904	53.88	6.00
Both	12,141	23.71	4.00
Neither	16,146	49.73	6.00
Regime transitions per ASIN			
Mean	34.92		
Median	26.00		

Notes: Coffee category; ASIN-day panel (midnight prices), January 2021–March 2025. ASINs are restricted to those satisfying the event-study screen used in this script: at least 10 consecutive days with both Amazon and the lowest 3P price (NEW_third) non-missing, and at least 10 consecutive days with only the lowest 3P price observed (and Amazon missing). Regimes are defined by price availability of Amazon and FBA. "3P only": 3P price observed and Amazon missing. "Amazon only": Amazon observed and 3P missing. "Both": both observed. "Neither": neither observed.

C.3 Additional Facts on Amazon-3P Relative Prices

To make a precise comparison of 3P sellers' and Amazon's prices in product-weeks in which both are selling, Tables C.6 to C.9 display the results from a regression of logged prices on an indicator for Amazon, with ASIN and year-week fixed effects. It becomes apparent that, if anything, Amazon tends to set a lower price than 3P sellers, especially in the coffee category.

Table C.6: Price differences for identical products: Coffee

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	-0.2652***	-0.1860***	-0.4245***	-0.2609***	-0.1910***	-0.4266***
	(0.0072)	(0.0082)	(0.0069)	(0.0078)	(0.0088)	(0.0075)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	286,808	202,772	273,438	247,398	177,311	236,672
Adjusted R^2	0.88493	0.92332	0.91610	0.88489	0.92270	0.91726

Notes: The unit of observation is an ASIN–week-seller type in the Coffee category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one 3P seller of the relevant type (any 3P, lowest FBA, or lowest FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding 3P offer for the same product in the same week. "Popular products" are defined in Appendix B. Standard errors clustered by ASIN in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table C.7: Price differences for identical products: Cocoa

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	-0.0395	-0.0568*	-0.2676***	-0.0339	-0.0449	-0.2808***
	(0.0469)	(0.0306)	(0.0351)	(0.0529)	(0.0319)	(0.0382)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	13,504	10,371	12,917	11,730	9,195	11,231
Adjusted R^2	0.90274	0.93924	0.93063	0.87096	0.91966	0.90680

Notes: The unit of observation is an ASIN–week-seller type in the Cocoa category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one 3P seller of the relevant type (any 3P, lowest FBA, or lowest FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding 3P offer for the same product in the same week. "Popular products" are defined in Appendix B. Standard errors clustered by ASIN in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table C.8: Price differences for identical products: SD Cards

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	-0.0309	-0.0006	-0.1016***	-0.0150	0.0578	-0.0895**
	(0.0192)	(0.0243)	(0.0226)	(0.0298)	(0.0376)	(0.0360)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	20,218	15,719	19,126	10,620	7,963	9,968
Adjusted R^2	0.91216	0.92688	0.91114	0.89789	0.91437	0.89695

Notes: The unit of observation is an ASIN–week-seller type in the SD Cards category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one 3P seller of the relevant type (any 3P, lowest FBA, or lowest FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding 3P offer for the same product in the same week. "Popular products" are defined in Appendix B. Standard errors clustered by ASIN in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table C.9: Price differences for identical products: USB Hubs

	All products			Popular products		
Dependent variable	log(price)					
Compared to	Lowest 3P	FBA	FBM	Lowest 3P	FBA	FBM
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1{Amazon}	0.0028	0.0289	-0.0914***	0.0046	0.0542	-0.0967***
	(0.0170)	(0.0211)	(0.0146)	(0.0214)	(0.0336)	(0.0178)
ASIN FE	✓	✓	✓	✓	✓	✓
Week FE	✓	✓	✓	✓	✓	✓
Observations	36,666	27,215	31,872	19,786	14,702	16,881
Adjusted R^2	0.92494	0.93599	0.93954	0.92581	0.93043	0.94510

Notes: The unit of observation is an ASIN–week-seller type in the USB Hubs category. The dependent variable is the log of the posted price. For each column, we restrict the sample to ASIN–week observations in which both Amazon and at least one 3P seller of the relevant type (any 3P, lowest FBA, or lowest FBM) are active. The coefficient on 1{Amazon} measures the average log price difference between Amazon and the corresponding 3P offer for the same product in the same week. "Popular products" are defined in Appendix B. Standard errors clustered by ASIN in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

D Additional Evidence on Amazon and 3P Entry and Exit

D.1 Pre-trend Diagnostics

Table D.1: Pre-trend Wald tests: joint significance of pre-event coefficients, Amazon entry/exit event studies

Outcome	Event	Restrictions	F-stat	df1	df2	p-value
log(FBA)	Entry	9	0.887	9	413.0	0.5366
log(FBM)	Entry	9	0.539	9	604.0	0.8467
log(FBA)	Exit	9	1.337	9	353.0	0.2162
log(FBM)	Exit	9	1.457	9	488.0	0.1610

Notes: H0: all pre-event coefficients (entry: t = -pre_window, ..., -2; exit: t = -exit_pre_window, ..., -2) are jointly zero; a rejection would signal pre-existing differential trends. F-tests use fixest::wald(), which respects the two-way clustered (ASIN x date) covariance used in each event-study model.

Table D.2: Pre-trend Wald tests: joint significance of pre-event coefficients, stacked event studies (main price outcomes: log(FBA), log(FBM))

Outcome	Event	Path	Restrictions	F-stat	df1	df2	p-value
log(FBA)	Entry	3P	9	1.237	9	502.0	0.2695
log(FBA)	Entry	Amazon - 3P	9	1.606	9	502.0	0.1105
log(FBA)	Entry	Amazon (implied)	9	1.069	9	502.0	0.3844
log(FBA)	Exit	3P	9	4.763	9	560.0	0.0000***
log(FBA)	Exit	Amazon - 3P	9	4.229	9	560.0	0.0000***
log(FBA)	Exit	Amazon (implied)	9	0.565	9	560.0	0.8258
log(FBM)	Entry	3P	9	1.478	9	683.0	0.1520
log(FBM)	Entry	Amazon - 3P	9	0.868	9	683.0	0.5534
log(FBM)	Entry	Amazon (implied)	9	0.507	9	683.0	0.8699
log(FBM)	Exit	3P	9	0.611	9	703.0	0.7885
log(FBM)	Exit	Amazon - 3P	9	0.365	9	703.0	0.9514
log(FBM)	Exit	Amazon (implied)	9	0.518	9	703.0	0.8624

Notes: H0: all pre-event coefficients (t = -10, ..., -2, relative to the omitted t = -1) are jointly zero for the indicated path. 3P = third-party event-time path; Amazon − 3P = differential path; Amazon (implied) = 3P + differential. F-tests use fixest::wald(), which respects the two-way clustered (ASIN x date) covariance used in each stacked event-study model.

D.2 Additional Estimation Results

Table D.3: Event-studies outcomes: 3P lowest price under FBA or FBM

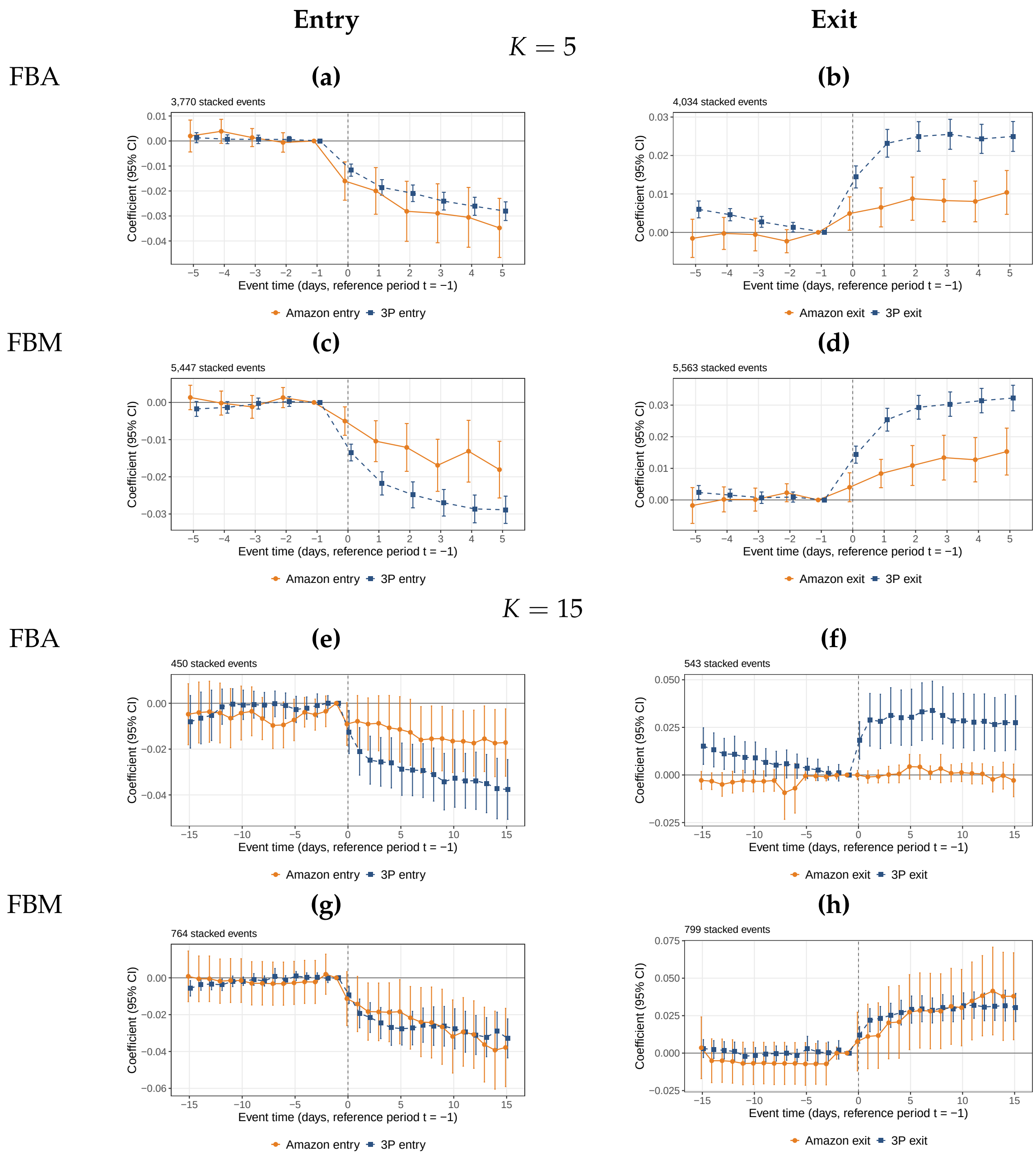
Dependent Variables:	$\log(p^{3P,FBA})$	$\log(p^{3P,FBM})$	$\log(p^{3P,FBA})$	$\log(p^{3P,FBM})$
Event:	Entry		Exit	
Model:	(1)	(2)	(3)	(4)
event_time = -10	0.0001	0.0038	0.0021	-0.0002
	(0.0048)	(0.0037)	(0.0040)	(0.0040)
event_time = -9	0.0007	0.0022	0.0026	3.69×10^{-5}
	(0.0046)	(0.0029)	(0.0041)	(0.0040)
event_time = -8	-0.0006	0.0033	0.0010	-0.0025
	(0.0045)	(0.0027)	(0.0045)	(0.0037)
event_time = -7	0.0005	0.0018	-0.0003	1.16×10^{-5}
	(0.0042)	(0.0026)	(0.0043)	(0.0036)
event_time = -6	7.58×10^{-5}	0.0009	0.0034	0.0076
	(0.0038)	(0.0025)	(0.0044)	(0.0050)
event_time = -5	0.0025	7.86×10^{-5}	0.0018	0.0006
	(0.0032)	(0.0024)	(0.0037)	(0.0040)
event_time = -4	0.0028	-0.0009	-0.0016	0.0051
	(0.0029)	(0.0026)	(0.0036)	(0.0033)
event_time = -3	0.0015	-0.0011	0.0017	0.0032
	(0.0027)	(0.0017)	(0.0032)	(0.0029)
event_time = -2	0.0013	-0.0009	-0.0013	0.0029
	(0.0023)	(0.0017)	(0.0024)	(0.0024)
event_time = 0	-0.0186***	-0.0086***	0.0106***	0.0069***
	(0.0038)	(0.0022)	(0.0033)	(0.0021)
event_time = 1	-0.0253***	-0.0153***	0.0156***	0.0099***
	(0.0042)	(0.0031)	(0.0038)	(0.0032)
event_time = 2	-0.0326***	-0.0181***	0.0183***	0.0192***
	(0.0045)	(0.0036)	(0.0041)	(0.0043)
event_time = 3	-0.0379***	-0.0189***	0.0242***	0.0257***
	(0.0049)	(0.0041)	(0.0046)	(0.0043)
event_time = 4	-0.0438***	-0.0204***	0.0256***	0.0273***
	(0.0054)	(0.0042)	(0.0048)	(0.0048)
event_time = 5	-0.0507***	-0.0230***	0.0288***	0.0320***
	(0.0058)	(0.0040)	(0.0050)	(0.0046)
event_time = 6	-0.0536***	-0.0247***	0.0286***	0.0312***
	(0.0058)	(0.0044)	(0.0053)	(0.0048)
event_time = 7	-0.0562***	-0.0260***	0.0290***	0.0322***
	(0.0060)	(0.0044)	(0.0051)	(0.0050)
event_time = 8	-0.0606***	-0.0299***	0.0265***	0.0330***
	(0.0063)	(0.0045)	(0.0061)	(0.0054)
event_time = 9	-0.0635***	-0.0283***	0.0267***	0.0333***
	(0.0067)	(0.0048)	(0.0060)	(0.0053)
event_time = 10	-0.0635***	-0.0295***	0.0287***	0.0349***
	(0.0068)	(0.0048)	(0.0061)	(0.0056)
Event	✓	✓	✓	✓
Observations	11,403	17,703	10,353	14,301
Number of Events	543	843	493	681
Number of ASINs	414	605	354	489
R^2	0.97950	0.98778	0.98561	0.98530

Notes: Two-way clustered (ASIN × date) standard errors in parentheses. Columns (1) and (2) report entry event studies; columns (3) and (4) report exit event studies. Dependent variables are log(lowest 3P seller price under FBA) in columns (1) and (3) and log(lowest 3P seller price under FBM) in columns (2) and (4). Coefficients are event-time indicators relative to $t = -1$ (omitted period), as specified in equation 3.

D.3 Robustness and Additional Outcomes

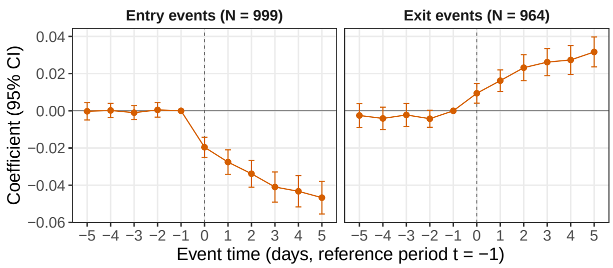
D.3.1 Alternative Qualifying-Spell Requirements

Figure D.1: Stacked event studies under alternative qualifying-event definitions

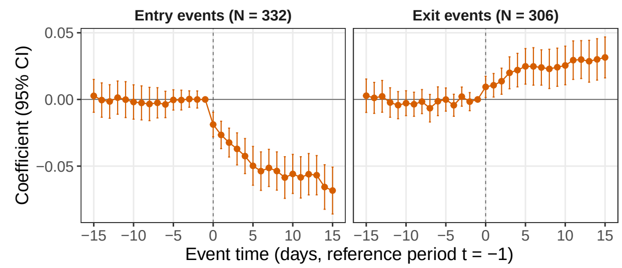


Notes: The figure reports stacked event-study estimates for Amazon and 3P seller entry and exit under alternative qualifying-event definitions. Panels (a)-(d) use $K = 5$, and panels (e)-(h) $K = 15$. Within each group, the top row outcome is the logarithm of the lowest 3P price under FBA, and the bottom-row under FBM. The left column reports entry events and the right column exit events. Relative to the baseline definition $K = 10$, $K = 5$ requires shorter pre- and post-event periods and admits a larger set of qualifying events, whereas $K = 15$ selects a smaller set of events with more persistent seller-presence regimes. The omitted event day is $r = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date. All regressions use the daily coffee data and otherwise follow Equation 4.

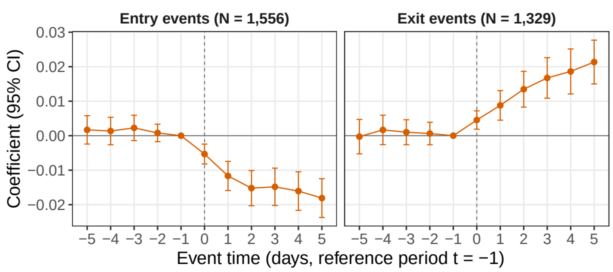
Figure D.2: Amazon entry and exit under alternative qualifying-event definitions



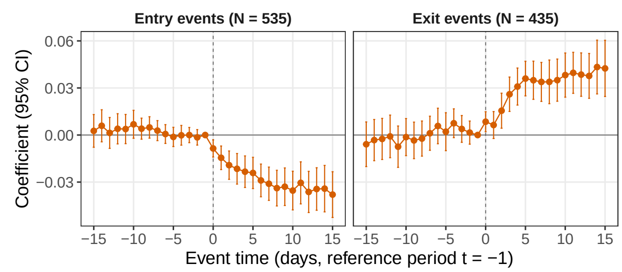
(a) $K = 5$, FBA.



(b) $K = 15$, FBA.



(c) $K = 5$, FBM.

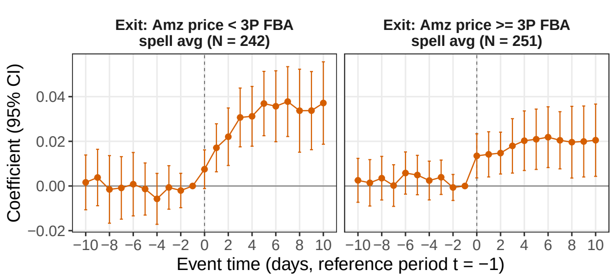


(d) $K = 15$, FBM.

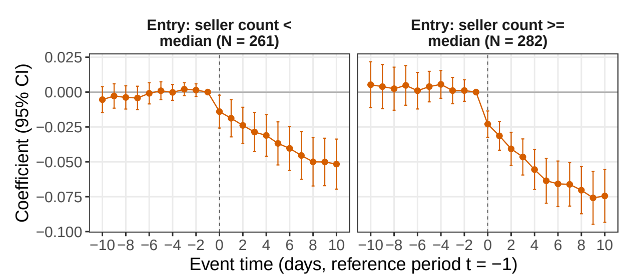
Notes: The figure reports Amazon entry and exit event-study estimates under alternative definitions of qualifying events. Panels (a) and (c) use $K = 5$, which admits events with shorter continuous pre- and post-transition regimes. Panels (b) and (d) use $K = 15$, which imposes a more restrictive event definition requiring longer continuous regimes around the transition. Panels (a) and (b) use the logarithm of the lowest FBA price as the outcome, while panels (c) and (d) use the logarithm of the lowest FBM price. The baseline specification uses $K = 10$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date.

D.3.2 Sample splits and response heterogeneity

Figure D.3: Heterogeneity in 3P price responses



(a) Amazon exit, split by Amazon's relative price.

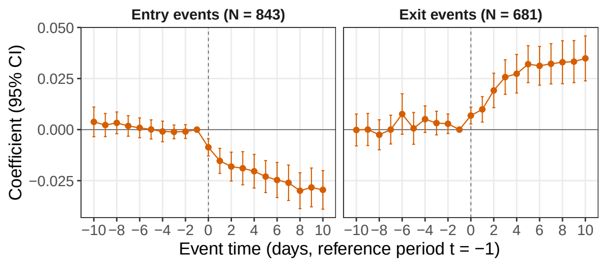


(b) Amazon entry, split by seller counts.

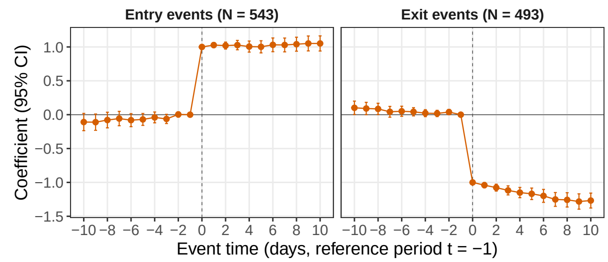
Notes: Panel (a) reports Amazon exit event-study estimates for subsamples defined by whether Amazon's price at entry is below versus at or above the pre-period lowest 3P price under FBA. The outcome is the logarithm of the lowest FBA price. The omitted event day is $r = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date. See the text for the precise definition of the subsamples. Panel (b) reports event-study estimates from Equation 3 for the logarithm of the lowest 3P FBA price, separately by whether the pre-period total seller count is below, or greater or equal to the median seller count among the products considered. The omitted event day is $r = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date.

D.3.3 Additional market outcomes

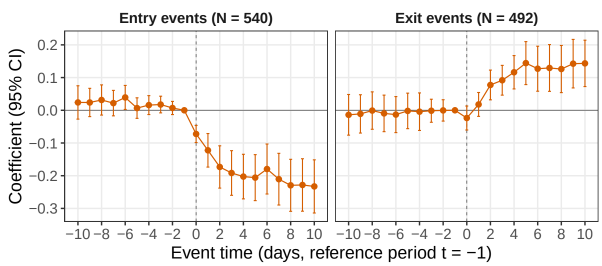
Figure D.4: Amazon entry and exit: additional event-study outcomes



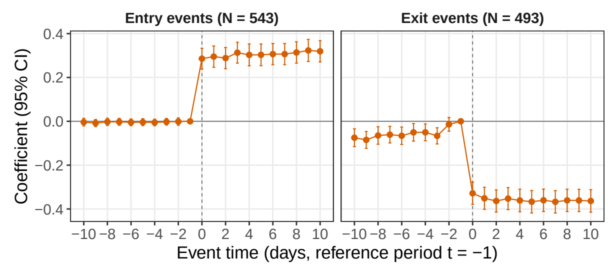
(a) Outcome: $\log(p^{3P,FBM})$.



(b) Outcome: seller count.



(c) Outcome: $\log(\text{sales rank})$.



(d) Outcome: $\mathbb{1}\{\text{Amazon wins the Buy Box}\}$.

Notes: The figure reports event-study estimates from Equation 3 using outcomes other than the baseline lowest FBA price. Panel (a) uses the logarithm of the lowest FBM price, panel (b) uses the total number of marketplace sellers, panel (c) uses the logarithm of sales rank, and panel (d) uses an indicator for whether Amazon occupies the Buy Box. The omitted event day is $r = -1$. Vertical bars report 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar date.

D.3.4 Alternative specifications

Tables D.4 and D.5 show the results from a more naive, but more standard, two-way fixed effects (TWFE) specification using data at the daily level. We run separate regressions for each product category. For product i observed on day t, we estimate:

$$\log\left(p_{it}^{3P,m}\right) = \beta \cdot \mathbb{I}(\text{Amazon}_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{9}$$

where, as in the main text, $p_{it}^{3P,m}$ denotes the lowest price offered by 3P sellers under $m \in \{FBA, FBM\}$ for ASIN i on day t. $\mathbb{I}(\text{Amazon}_{it})$ is an indicator taking the value 1 if Amazon is active as a seller for product i at time t. The inclusion of product fixed effects (μ_i) and time fixed effects (λ_t) absorbs time-invariant product characteristics (e.g., quality) and common macro-shocks, respectively.

Table D.4: Amazon's presence and 3P lowest price (FBA), by product category

Product	Coffee	Cocoa	Pasta	Shirts	TVs	USBhubs
Dependent Variable:			$\log(p^{3P,FBA})$			
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1(Amazon present)	-0.092***	-0.073***	-0.108***	-0.070***	-0.028***	-0.076***
	(0.006)	(0.020)	(0.011)	(0.010)	(0.008)	(0.020)
asin	✓	✓	✓	✓	✓	✓
yearmonthday	✓	✓	✓	✓	✓	✓
Observations	929,797	66,371	394,602	96,467	116,668	124,066
N ASINs	1,411	82	590	206	386	172
R^2	0.90951	0.94421	0.92521	0.77564	0.98214	0.94066

Notes: Standard errors clustered by ASIN in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$. All columns cover the sample period January 2021 to March 2025. Every column is further restricted to ASINs with ≥10 consecutive days with Amazon and 3P both priced, and ≥10 consecutive days with 3P priced while Amazon is absent.

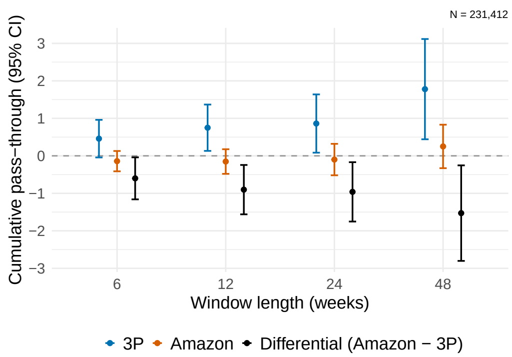
Table D.5: Amazon's presence and 3P lowest price (FBM), by product category

Product	Coffee	Cocoa	Pasta	Shirts	TVs	USBhubs
Dependent Variable:			$\log(p^{3P,FBM})$			
Model:	(1)	(2)	(3)	(4)	(5)	(6)
1(Amazon present)	-0.030***	-0.054***	-0.048***	-0.020**	-0.048***	-0.067***
	(0.006)	(0.018)	(0.008)	(0.009)	(0.007)	(0.016)
asin	✓	✓	✓	✓	✓	✓
yearmonthday	✓	✓	✓	✓	✓	✓
Observations	1,487,611	87,099	641,785	182,797	367,899	111,807
N ASINs	1,601	89	746	243	497	131
R^2	0.91264	0.95081	0.91872	0.80469	0.96714	0.91209

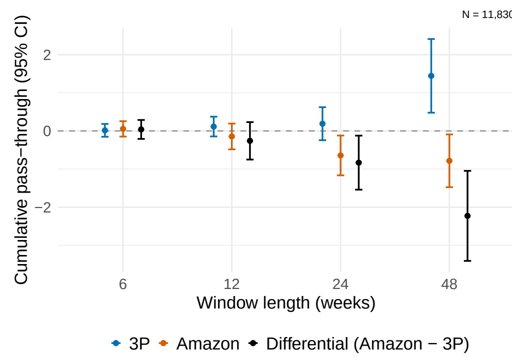
Notes: Standard errors clustered by ASIN in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$. All columns cover the sample period January 2021 to March 2025. Every column is further restricted to ASINs with ≥10 consecutive days with Amazon and 3P both priced, and ≥10 consecutive days with 3P priced while Amazon is absent.

E Asymmetric Cost Pass-Through: Additional Specifications

Figure E.1: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample excluding high-priced ASINs



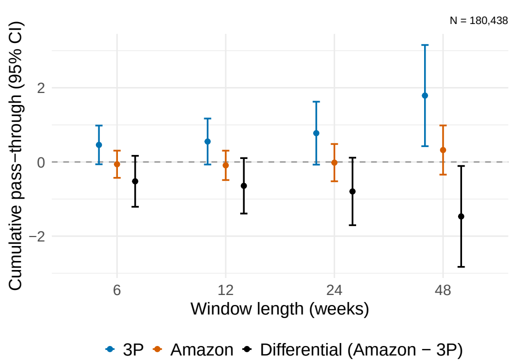
(a) Coffee.



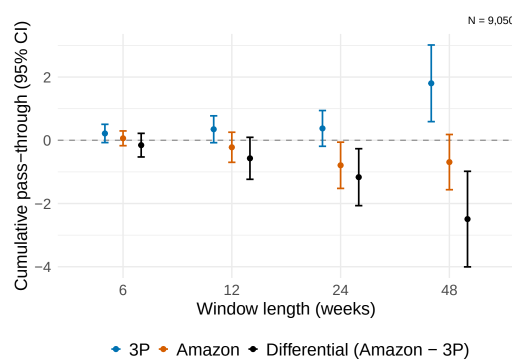
(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include ASIN $\times$ seller-type fixed effects. The sample additionally excludes ASINs whose median price exceeds the 80th percentile of prices in the category. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.2: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample excluding ASINs without $\geq$ 48 week consecutive presence by both seller types



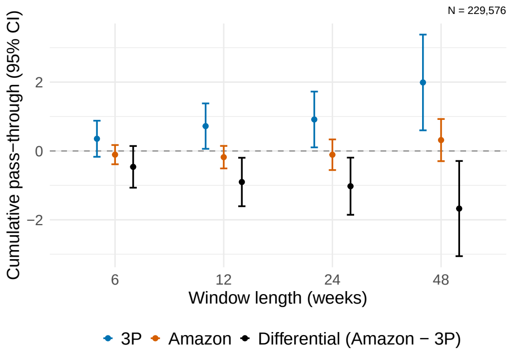
(a) Coffee.



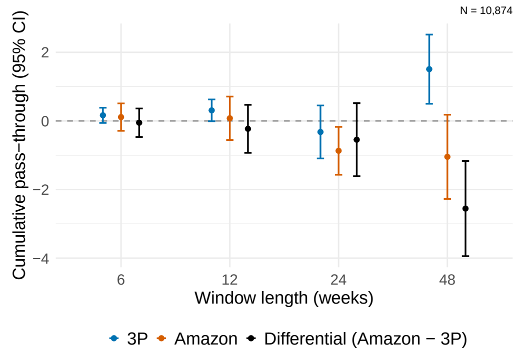
(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include ASIN $\times$ seller-type fixed effects. The sample excludes ASINs that do not have at least 48 weeks of consecutive presence by both seller types. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.3: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample, "popular" (higher-turnover) ASINs only



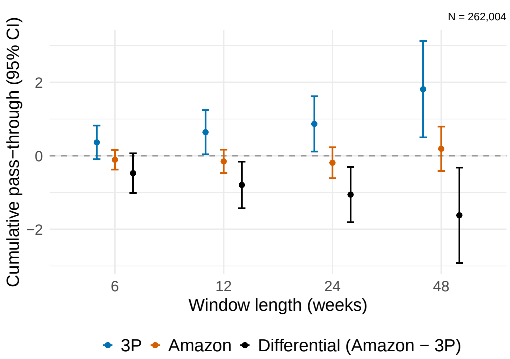
(a) Coffee.



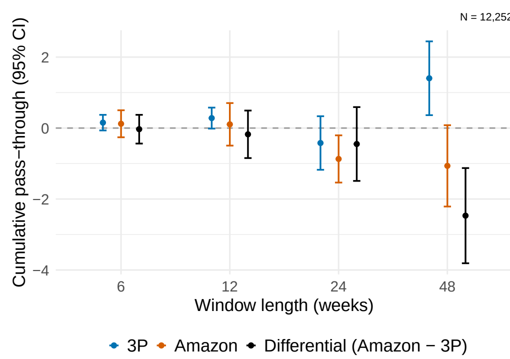
(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include ASIN $\times$ seller-type fixed effects. This sample only uses ASINs that are "popular", as defined in Appendix B. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.4: Cumulative commodity-cost pass-through by seller type: matched ASIN-week sample excluding ASINs without reviews



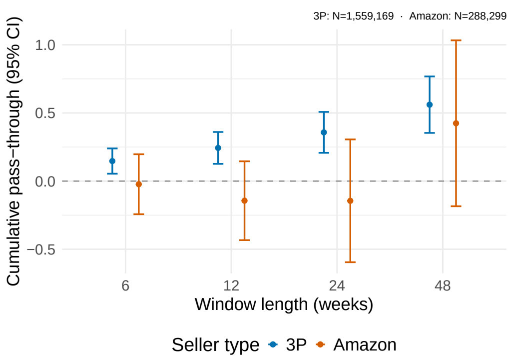
(a) Coffee.



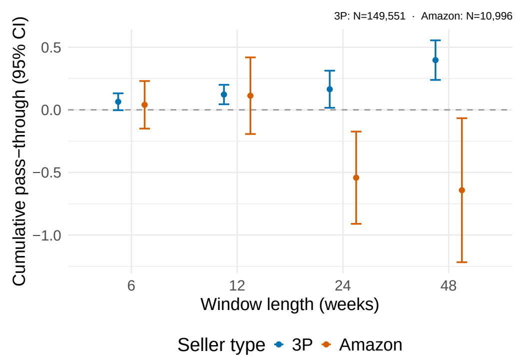
(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions use a matched sample of ASIN-week cells with observed weekly price changes for both Amazon's posted price and the lowest 3P price, and include ASIN $\times$ seller-type fixed effects. The sample additionally excludes ASINs that do not have any user review. "Differential (Amazon - 3P)" is the difference between the two estimates; negative values indicate lower pass-through by Amazon. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.5: Cumulative commodity-cost pass-through by seller type: seller-specific ASIN-week samples



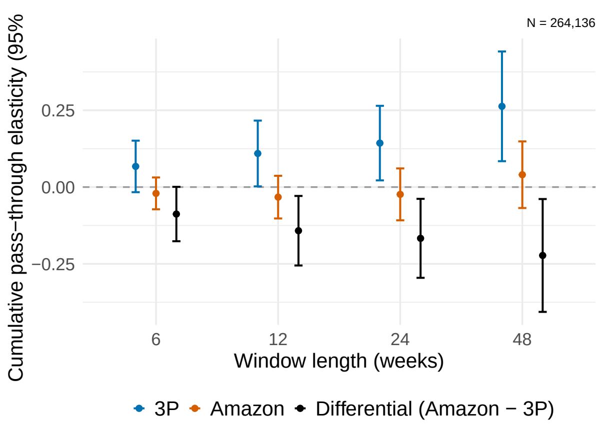
(a) Coffee.



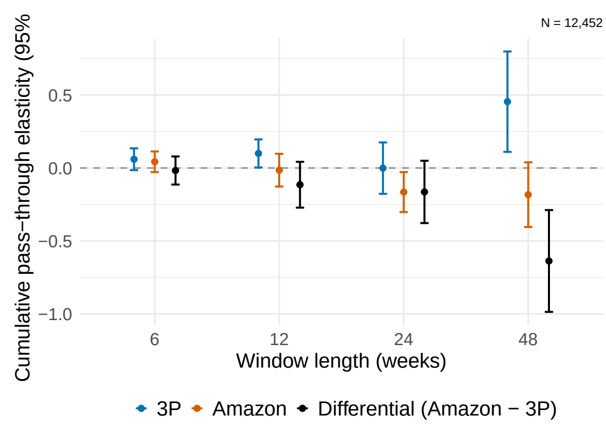
(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions are estimated separately for Amazon's posted price and the lowest 3P price, using all ASIN-week cells with an observed weekly price change for the relevant seller type and including ASIN fixed effects. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.6: Commodity-cost pass-through in logarithms



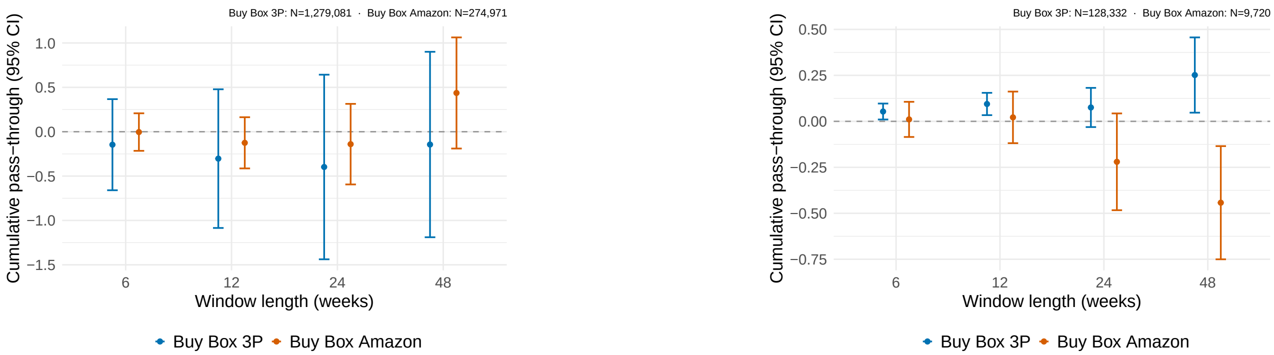
(a) Coffee.



(b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from the logarithmic counterpart of the baseline stacked common-support specification. Regressions relate weekly log changes in posted retail prices to weekly log changes in the relevant commodity-cost series. The Amazon and lowest-3P price responses are estimated on the same set of ASIN-week cells. "Differential (Amazon − 3P)" denotes the difference between the cumulative Amazon and 3P estimates. Vertical bars report 95intervals based on standard errors two-way clustered by ASIN and calendar week.

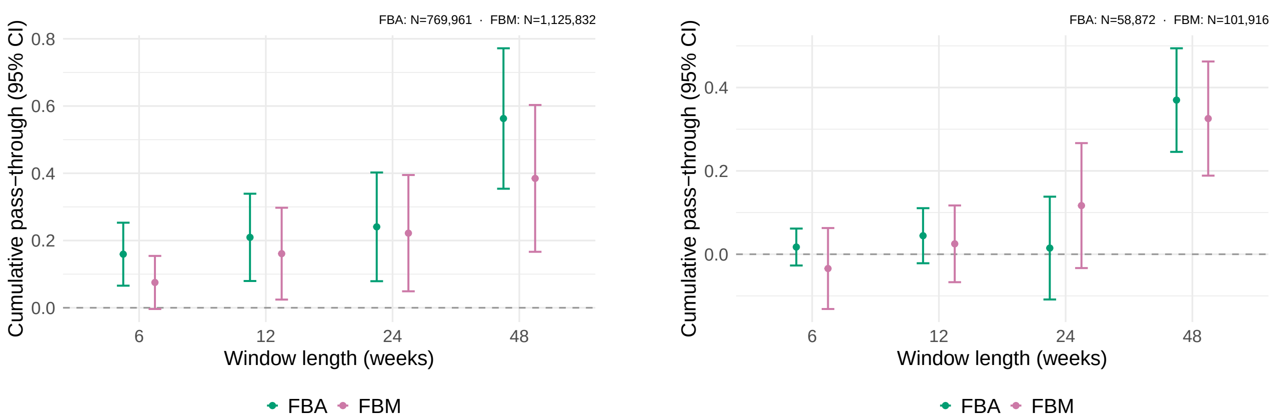
Figure E.7: Cumulative commodity-cost pass-through using Buy Box prices: seller-specific ASIN-week samples



(a) Coffee. (b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions are estimated separately by seller type using Buy Box prices in ASIN-weeks in which that seller type is displayed in the Buy Box, and include ASIN fixed effects. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week.

Figure E.8: Cumulative commodity-cost pass-through for 3P prices by fulfillment mode



(a) Coffee. (b) Cocoa.

Notes: The figure reports cumulative pass-through estimates from distributed-lag regressions of weekly changes in posted retail prices on weekly changes in converted commodity costs. At window length K, the contemporaneous coefficient and first $K-1$ lag coefficients are summed. Both variables are measured in dollars per gram, so an estimate of one denotes full pass-through. Regressions are estimated separately for the lowest 3P prices under FBA and FBM, using all ASIN-week cells with an observed weekly price change for the relevant fulfillment mode and including ASIN fixed effects. Vertical bars show 95% confidence intervals based on standard errors two-way clustered by ASIN and calendar week. The finding is that the pass-through point estimates have a tendency to be slightly higher for 3P sellers operating under FBA, as opposed to sellers operating under FBM, with the exception of the Cocoa estimates at 24 lags.